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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

ZINC NICKEL

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "what is an alloy?" to a signed certificate of completion. Alkaline zinc-nickel (12–16% Ni) on steel, the corrosion-protection heavyweight. Assumes you know nothing. Teaches you everything.

01 CLEAN ▸ 02 RINSE ▸ 03 ACTIVATE ▸ 04 RINSE

05 PLATE ZNNI ▸ 06 RINSE ▸ 07 ACTIVATE ▸ 08 PASSIVATE ▸ 09 SEAL/DRY

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — a **zinc-nickel alloy plating line**. Maybe you started this week. Maybe you've been racking parts for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here's the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, alloys, or electricity.** That isn't an insult. That's a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what "alloy codeposition" means or why a thirty-second rinse can decide whether a brake caliper survives ten years on a salty road. You learn it. We're going to teach it.

This manual is written in the spirit of the *plain-language* guides — friendly, plain-spoken, and patient. We'd rather over-explain than leave you guessing in front of a tank full of hot caustic and nickel.

> ★ **REMEMBER**
> You don't need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it, rinse it, activate it again, passivate it, then seal and dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you will run into little flagged notes. Here is what each one means.

> 💡 **TIP**
> A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.

> ★ **REMEMBER**
> A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because a zinc-nickel line has real hazards — especially the hot caustic, the nickel, and the amine complexors at its heart.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.

> ☀ **FUN FACT**
> A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

> 💡 **TIP**
> Read the Fundamentals all the way through *before* you read any individual station chapter. The station chapters assume you already understand the big picture this section gives you. It's the difference between reading a recipe when you already know how to cook versus the very first time you've stepped into a kitchen.

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist (sometimes paired with a red "Top Mistakes" card) — the things you must be able to *say out loud, unprompted* before you're cleared on that station; it's how you and your trainer confirm the ideas actually stuck. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU’LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal onto a part.
2. **Describe what zinc-nickel alloy plating does** and why a shop chooses it — *outstanding* corrosion protection, thermal stability, and a tough alloy finish that far outlasts plain zinc.
3. **Explain what an alloy deposit is** and why zinc and nickel plate *together*, and why the **12–16% nickel** range is the “sweet spot.”
4. **Name all nine stages of the line in order** and explain *why* the order cannot be rearranged.
5. **Understand what each tank is for** — what goes in, what comes out, and what “good” looks like at every handoff.
6. **Recognize the main hazards** at each station — hot caustic, nickel salts, amine complexors, mineral acids, electrical, mist, and chromate chemistry — and know the correct PPE and emergency response for each.
7. **Use the basic field tests** every operator relies on — the water-break test, conductivity checks, and reading an **XRF %Ni** result.
8. **Spot trouble early** — read a deposit (and an alloy reading) and know whether the problem is upstream, in the bath, in the alloy balance, or in your handling.
9. **Understand hydrogen embrittlement** and why high-strength steel **must be baked.**
10. **Talk the language** — use the right words with your lead, your chemist, and your chemistry supplier.

> ★ **REMEMBER**
> You’re not expected to hit all of these on day one. Plating is a craft. Zinc-nickel is one of the more demanding versions of that craft. The goal is steady competence, not instant mastery.

• THE NINE-STAGE PROCESS AT A GLANCE

Here’s the whole zinc-nickel line in one strip. Don’t memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Rinse → Plate (ZnNi) → Rinse → Activate → Passivate → Seal/Dry.**

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows almost every working tank.** Rinses aren’t breaks — they’re checkpoints that keep one chemistry from poisoning the next. Every working chemistry wants to “hitch a ride” into the next tank on the surface of your part, and rinses are what stop that. They aren’t filler between the “real” steps; they’re some of the most important steps you’ll perform.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. With zinc-nickel, there's a twist: we're depositing **two metals at once** — mostly zinc, with a carefully controlled slice of nickel mixed in. That mixture is an **alloy**, and it's what gives this finish its superpowers. We'll get to the alloy part in Chapter 3. First, the basics.

Imagine you have a steel bracket. Steel is strong and cheap, but it rusts. You want to give it a thin protective skin — and for the harshest jobs (under a truck, in an engine bay, on a brake component), plain zinc isn't tough enough. A **zinc-nickel alloy** coating protects far longer and survives heat far better. We use electricity to grow that alloy directly onto the steel, atom by atom. That process is electroplating.

HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

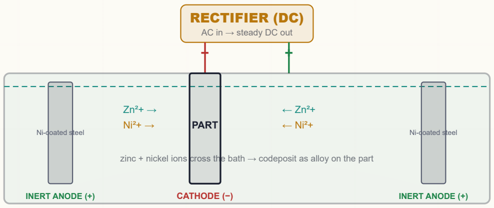
Picture a tank full of liquid called the **electrolyte** (or just "the bath"). Dissolved in it are **zinc** and **nickel**, both broken down into tiny charged particles called **ions**. An *ion* is just an atom carrying an electrical charge. Zinc floats around as **Zn^{2+}**; nickel floats around as **Ni^{2+}**.

You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current plating needs):

- **The cathode** — this is your *part* (your steel bracket), connected to the **negative** terminal. *Cathode = the part = the thing collecting metal.*
- **The anode** — the positive side. Here zinc-nickel does something different from acid zinc, and you need to know it now: **the anodes in an alkaline zinc-nickel bath are usually NOT made of zinc-nickel.** Most modern lines use **inert (insoluble) anodes** — often nickel-coated steel or specialized coated anodes — and feed the zinc and nickel into the bath *separately*. More on this special setup at Station 05.

When you turn on the rectifier:

1. Current flows. The positively-charged zinc and nickel ions are *attracted* to the negatively-charged part, because opposite charges attract.
2. When the ions reach the part's surface, they grab electrons and turn back into solid metal — zinc and nickel together — building up the alloy coating.
3. At the anodes, the chemistry is more complicated than in acid zinc (the anodes don't simply dissolve to refill the metal), which is exactly why zinc-nickel needs more active chemistry management. We'll explain at Station 05.



★

REMEMBER

Three words to lock in now: **Cathode = the part** (collects the alloy). **Anode = the (usually inert) bars. Electrolyte = the bath** (the caustic liquid carrying both zinc and nickel). Almost everything in plating is a variation on this one idea — zinc-nickel just adds a second metal to the trip.

⚙

TECHNICAL STUFF

At the part, two reductions happen side by side: $Zn^{2+} + 2e^- \rightarrow Zn^0$ and $Ni^{2+} + 2e^- \rightarrow Ni^0$, codepositing as an alloy. A *third* reaction also runs hard here — water splitting to make hydrogen gas ($2H_2O + 2e^- \rightarrow H_2 + 2OH^-$) — which is why alkaline zinc-nickel is a **low-efficiency** bath (lots of current makes hydrogen, not metal). That same hydrogen is the reason high-strength parts must be baked.

☀ **FUN FACT**

“Galvanize” and the unit “volt” trace back to Luigi Galvani and Alessandro Volta, two 1790s Italian scientists who argued over whether electricity came from frog legs or from metals. Volta won — and you’ll read volts on your rectifier every day.

⚠ **HEADS-UP**

Plating rectifiers push enormous current — hundreds to thousands of amps — low voltage but very high current, into a tank full of conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus hot caustic plus your hand demands total respect.

WHY TWO METALS INSTEAD OF ONE?

You might reasonably ask: if plain zinc already protects steel, why bother adding nickel and all the complications that come with it? The short answer is **endurance.** Plain zinc is a fine, fast, cheap rust-protector for ordinary parts. But the world has parts that live a hard life — bolts on a truck chassis sprayed with road salt all winter, brackets in an engine bay that bake and cool a thousand times, brake hardware that has to survive a decade of weather. For those, plain zinc wears out too soon.

Mixing in a small, controlled amount of nickel — just 12 to 16 percent — transforms the coating. The alloy corrodes far more slowly and far more evenly than plain zinc, and it keeps protecting even after heat that would ruin a plain-zinc finish. You spend more time and more money to make it, and in return you get a coating that lasts five to ten times longer in a salt-spray test. That trade — more cost and care for dramatically more life — is the entire reason this line exists.

💡 **TIP**

Whenever the work feels fussier than plain zinc — slower plating, more chemistry to watch, an extra activation and a seal at the end — remember *why*. Every bit of that extra care is buying corrosion life the customer specifically paid for. You’re not running a slow zinc line; you’re running a premium-corrosion line.

★ **REMEMBER**

Electroplating moves metal from a bath onto your part using electric current. Zinc-nickel does it with *two* metals at once — mostly zinc, a measured slice of nickel — to make an alloy that outlasts plain zinc by a wide margin. Hold that one idea and the rest of this manual is just detail.

CHAPTER 2

WHAT IS "ZINC-NICKEL"?

AND WHY DO WE USE IT — THE CORROSION-PROTECTION HEAVYWEIGHT OF THE ZINC FAMILY

ZINC PLATING COMES IN FAMILIES

"Zinc plating" just means coating something with zinc. But there are *several recipes* for the bath that does it, and zinc-nickel is the high-performance member of the family:

- **Acid zinc** — an acidic chloride bath. Fast and bright, but ordinary corrosion protection.
- **Alkaline zinc** — a high-pH (caustic) bath. Great even coverage, ordinary corrosion protection.
- **Zinc-nickel** — an alloy bath (this manual). It deposits zinc *plus* about **12–16% nickel**, delivering **dramatically better corrosion protection and heat resistance** than any plain-zinc process. It's what you reach for when "regular zinc isn't good enough."

Zinc-nickel itself comes in two sub-families:

- **Alkaline zinc-nickel** (high pH, caustic-based) — **the dominant industrial process and the focus of this manual.** Excellent alloy uniformity (consistent %Ni across the whole part), best for the automotive and fastener specs that demand it.
- **Acid (chloride) zinc-nickel** (low pH) — faster and a bit more efficient, often used for wire, strip, and some barrel work, but the alloy %Ni tends to vary more across the part. We'll point out the differences where they matter.

> ⚙ **TECHNICAL STUFF**
> **pH** is the scale for how acidic or basic a liquid is, 0 (strong acid) to 14 (strong base), 7 neutral. Our alkaline zinc-nickel bath runs *very* basic — typically **pH 13–14** — because it's built on sodium hydroxide (caustic soda). That's the opposite end of the scale from acid zinc's pH ~5. High pH is a real burn hazard; respect it.

More precisely, our process is **alkaline (caustic) zinc-nickel.** The supporting chemistry is **sodium hydroxide (NaOH, "caustic soda")**, which dissolves the zinc as **zincate** and carries the current. The nickel can't just be tossed in — at high pH it would precipitate as a useless sludge — so it's held in solution by **amine complexors** (organic molecules that "wrap around" the nickel and keep it dissolved and available to plate). A package of **brighteners and additives** controls grain, appearance, and the all-important alloy uniformity.

THE BATH AT A GLANCE (ALKALINE ZINC-NICKEL)

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Zinc metal (as zincate)	`6–12 g/L (0.8–1.6 oz/gal)`	One half of the alloy you're plating
Nickel metal (complexed)	`0.6–1.6 g/L (0.08–0.21 oz/gal)`	The second half — the "nickel" in zinc-nickel
Sodium hydroxide (NaOH)	`100–140 g/L (13–19 oz/gal)`	Dissolves zinc as zincate; carries current
Zn : Ni ratio *in the bath*	`~8:1 to 12:1 (Zn to Ni)`	A master control for alloy %Ni and uniformity
Amine complexor system	`Per supplier's TDS`	Keeps nickel dissolved at high pH; controls codeposition
Brightener / additive	`Per supplier's TDS`	Grain refinement, brightness, alloy distribution
Carbonate (Na_2CO_3)	`< 60–90 g/L (watch)`	Builds up over time; eventually must be removed
pH	`13–14 (essentially "all caustic")`	Governed by NaOH level, not adjusted like acid baths

> ★ **REMEMBER**
> Read that table twice and notice how *little* metal is in this bath compared with acid zinc (which runs 15–45 g/L zinc). Alkaline zinc-nickel is a **low-metal, high-caustic** bath. The numbers that rule it aren't the absolute metal levels — they're the **Zn:Ni ratio** and the **caustic level**, because those two control the alloy you actually deposit.

TECHNICAL STUFF

"g/L" means grams per liter; "oz/gal" is the same idea in ounces per gallon. A **TDS** is a **Technical Data Sheet** — the supplier's instruction document for each product. When this manual says "per TDS," follow the supplier's sheet, because complexor and brightener levels are specific to each product line. Never guess at additive levels — in zinc-nickel they directly move the alloy composition.

HEADS-UP

The amine complexors that make this bath possible are also a **waste-treatment and exposure concern.** Some complexors are hard to break down in wastewater and can re-dissolve metals you're trying to remove. Treat the bath, its mist, and its rinses with the same respect as the caustic itself. Full PPE, good ventilation, never bare skin.

• WHY DO SHOPS USE ZINC-NICKEL?

Three big reasons: **outstanding corrosion protection**, **thermal stability**, and **a tough, uniform alloy** — the things plain zinc *can't* give you. Let's take them one at a time.

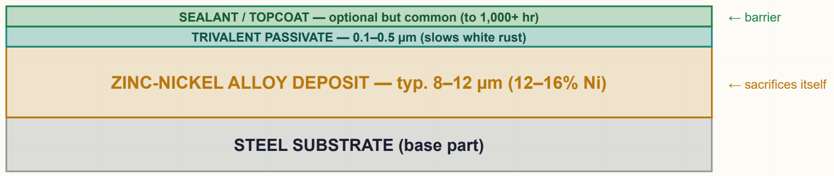
REASON 1: OUTSTANDING CORROSION PROTECTION (THE HEADLINE)

Like plain zinc, zinc-nickel protects steel **sacrificially** — the coating corrodes preferentially so the steel underneath survives, even at a scratch. But zinc-nickel does it *far* longer. The nickel in the alloy makes the coating corrode much more slowly and *evenly*, so the protection lasts.

The numbers tell the story. With a good **passivate plus a topcoat/seal**, zinc-nickel routinely hits **720 to 1,000+ hours of neutral salt spray before red rust** (ASTM B117) — versus roughly **96–240 hours** for passivated plain zinc. That's why zinc-nickel owns the demanding corrosion specs: automotive fasteners and underbody parts, brake components, engine-bay hardware, and military/aerospace work.

THE FINISH, LAYER BY LAYER (NOT TO SCALE)

ZnNi PROTECTS STEEL · PASSIVATE + SEAL PROTECT THE ZnNi



REMEMBER

Zinc-nickel protects steel by sacrificing itself — slowly, evenly, and for a long time. That's the entire reason this line exists. The nickel doesn't make the coating "stainless"; it makes the sacrifice last. Everything you do on this line is in service of laying down a good, even, correctly-alloyed layer.

REASON 2: THERMAL STABILITY

Plain zinc's protection falls off after exposure to heat (the coating and especially its chromate skin degrade). Zinc-nickel keeps protecting even after baking and after service heat — a big reason it's used in **engine and brake** areas where parts get hot. After a typical embrittlement bake, zinc-nickel retains far more of its corrosion performance than plain zinc would.

☀ **FUN FACT**

Zinc-nickel's heat tolerance is why you'll often see it on parts that live near hot machinery, while plain zinc gets the parts that stay cool. Two coatings made of mostly the same metal — but the 12–16% nickel completely changes where each one belongs.

REASON 3: A TOUGH, UNIFORM, HARD ALLOY

The zinc-nickel deposit is **harder and more wear-resistant** than plain zinc, and the alkaline process plates it **uniformly** — consistent thickness *and* consistent %Ni across the whole part, even into recesses. Uniform alloy composition is the whole game, which is why Chapter 3 and Part Four are devoted to controlling and measuring it.

THE 12–16% NICKEL "SWEET SPOT" — WHY IT MATTERS SO MUCH

Here's the single most important idea that makes zinc-nickel *zinc-nickel*: **the corrosion performance depends on getting the nickel content into a narrow window, roughly 12–16% (by weight), with ~13–15% as the bullseye.**

- **Too little nickel (below ~10%):** you lose the corrosion benefit — it starts behaving more like plain zinc.
- **Too much nickel (above ~18%):** the deposit can change phase, become less sacrificial (less protective of the steel), and develop stress/cracking and passivation problems.
- **In the 12–16% window:** the alloy forms the protective "gamma phase" structure that gives that legendary salt-spray life.

★ **REMEMBER**

More nickel is NOT better. Zinc-nickel has a *sweet spot*, not a "more is better" dial. Your job on this line is to keep the deposit *in the window*, every part, every shift. That's why we measure %Ni (with XRF — see Part Four) instead of just trusting the bath.

⚙ **TECHNICAL STUFF**

The protective structure in this window is the **gamma (γ) phase** of zinc-nickel, a specific crystal arrangement that's more corrosion-resistant and more uniform than the zinc-rich phases you get at low nickel. Drift out of the window and you start forming other phases that corrode differently — which is why composition control is treated as a quality-critical parameter, not a nicety.

☀ **FUN FACT**

People are sometimes surprised that *adding* an expensive metal like nickel and *only 12–16% of it* can multiply corrosion life five- to tenfold. It's not the amount — it's the alloy structure that small amount creates.

HOW THICK SHOULD THE COATING BE?

Zinc-nickel is usually specified by customer print or standard (e.g., **ASTM B841** covers zinc-nickel alloy coatings) and frequently by automotive specs. Typical commercial thickness runs **8–12 µm**, with severe-service parts at **12–20 µm**. As always, **the customer's spec decides** — your job is to plate to the called-out thickness *and* hit the %Ni window.

💡 **TIP**

You don't pick the thickness or the %Ni target — the customer's spec or print does. Before you start a job, confirm both the **thickness** and the **alloy %Ni** requirement on the traveler. A part that's the right thickness but the wrong alloy can still fail the spec.

THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **Low cathode efficiency (~50–65%, often lower).** A lot of current makes hydrogen gas instead of alloy, so zinc-nickel **plates slowly** — slower than plain alkaline zinc, and far slower than acid zinc. Plan for longer plate times.
- **Higher cost.** Nickel is expensive, the complexors are expensive, and the slow plating uses more energy and time. Zinc-nickel costs noticeably more than plain zinc — you use it when the corrosion spec *requires* it, not for everyday parts.
- **More complex chemistry to control.** You're now managing *two* metals and their ratio, the caustic, the complexors, *and* the resulting alloy composition. More knobs, more discipline.
- **Hydrogen embrittlement risk.** Like all low-efficiency caustic baths, it loads hydrogen into steel; high-strength parts **must be baked.**
- **Waste-treatment burden.** Nickel and amine complexors complicate effluent treatment (see Part Four).

> 💡 **TIP**
>
> If someone asks "why are we running zinc-nickel instead of regular zinc on this — it's slower and pricier," the answer is almost always one word: **corrosion.** The customer's spec calls for salt-spray life that plain zinc simply can't reach. That's not waste; that's the whole point of the process.

A WORD ABOUT THE FINAL STATIONS: PASSIVATE + SEAL

The zinc-nickel line ends with **trivalent chromate passivation** (a chemical conversion coating that protects the alloy and slows white rust) followed, very commonly, by a **seal or topcoat** that pushes salt-spray life into the 720–1,000+ hour range. Plain zinc lines often stop at passivate; **zinc-nickel usually adds the seal** because that's how it earns its premium corrosion numbers.

> ⚠ **HEADS-UP**
>
> Modern passivates use **trivalent chromium — Cr(III)** — the RoHS-compliant, far less hazardous form. The old generation used **hexavalent chromium — Cr(VI)** — a **confirmed human carcinogen**, heavily restricted (OSHA PEL 5 µg/m³, supplied-air respirators, medical surveillance per 29 CFR 1910.1026). **Know which one your shop uses.** Specialty zinc-nickel black passivates historically leaned on Cr(VI); make sure yours is trivalent unless management has signed off otherwise.

> ★ **REMEMBER**
>
> **Zinc-nickel without passivate and seal is leaving most of its corrosion performance on the table.** The alloy is the engine; the passivate and seal are the turbocharger that gets you to 1,000 hours.

CHAPTER 3

ALLOY CODEPOSITION BASICS

HOW TWO METALS PLATE AT ONCE — AND WHY IT'S A LITTLE STRANGE

This chapter exists because zinc-nickel is an **alloy** process, and alloys behave differently from single-metal plating. Spend ten minutes here and the whole rest of the manual makes sense.

WHAT "ALLOY" AND "CODEPOSITION" MEAN

An **alloy** is a mixture of metals — here, zinc and nickel blended at the atomic level, not layered. **Codeposition** means both metals plate **at the same time, onto the same part, out of the same bath.** When the rectifier is on, zinc ions *and* nickel ions both head for the part and lock in together, building one combined alloy coating.

THE SURPRISE: "ANOMALOUS" CODEPOSITION

Here's the weird, important part. You'd expect the *more noble* metal (nickel) to plate preferentially. In zinc-nickel, the **opposite** happens — the *less noble* metal (zinc) plates preferentially. So even though the bath might be, say, 90% zinc and 10% nickel by metal content, the **deposit** still comes out at only 12–16% nickel. Chemists call this **anomalous codeposition**, and it's actually a gift: it lets a low-nickel bath produce a controlled-nickel alloy.

> ⚙ **TECHNICAL STUFF**
> "Anomalous codeposition" means the alloy doesn't simply mirror the bath ratio — zinc suppresses nickel deposition at the part surface (tied to the local rise in pH and hydrogen evolution during plating). The practical upshot: small changes in **temperature, current density, and the Zn:Ni ratio** move the deposit's %Ni *more* than you'd guess. That sensitivity is exactly why we measure %Ni instead of assuming it.

> ★ **REMEMBER**
> **The bath ratio and the deposit ratio are NOT the same number.** A bath that's roughly 8:1 to 12:1 zinc-to-nickel yields a *deposit* that's only 12–16% nickel. Don't be alarmed that there's "so little nickel" in the tank — that's normal and correct.

THE THREE KNOBS THAT MOVE %NI

Because codeposition is sensitive, a handful of parameters quietly steer your alloy. You usually won't adjust these yourself (that's the chemist's job), but you must **recognize** them:

KNOB	TURN IT UP AND %NI IN THE DEPOSIT TENDS TO...	WHY IT MATTERS
Temperature	go **up** (more nickel codeposits when warmer)	Biggest everyday driver of alloy drift
Current density (CD)	go **down** at high CD; **up** at low CD	Edges vs. recesses can differ in %Ni
Zn:Ni bath ratio	more nickel in bath → more nickel in deposit	The chemist's main composition lever

> 💡 **TIP**
> If a job's %Ni reading is drifting and nobody changed chemistry, **check temperature first.** Temperature is the sneakiest cause of alloy drift on a zinc-nickel line — a tank running a few degrees hot all afternoon can walk the nickel right out of spec.

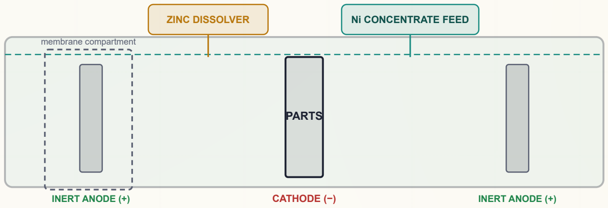
> ⚠ **HEADS-UP**
> Because %Ni changes with current density, the **edges and the recesses of a part can have slightly different alloy composition.** On critical parts this is checked with XRF at specified locations. Never assume one reading represents the whole part — measure where the spec tells you to.

WHY THIS BATH NEEDS INERT ANODES AND A SEPARATE NICKEL FEED

In acid zinc, zinc anodes dissolve to refill the metal automatically. **Zinc-nickel can't work that way**, for two reasons:

1. There's no good "zinc-nickel anode" that dissolves in the right 12–16% proportion — it would steadily throw off your alloy.
2. At high pH, you need tight, independent control of zinc *and* nickel.

So most alkaline zinc-nickel lines use **inert (insoluble) anodes** and feed the metals in *separately* — zinc from a side **zinc dissolver**, nickel from a **complexed nickel concentrate** dosed in. Some lines go further and put the anodes in a **membrane compartment** (a divided cell) to keep unwanted reactions away from the main bath. We cover this hardware at Station 05.



zinc + nickel fed separately · inert anodes carry current without throwing off the alloy

⚙ TECHNICAL STUFF — MEMBRANE (DIVIDED) CELLS

With inert anodes, an unwanted reaction can slowly *oxidize the amine complexors* at the anode, degrading the bath and hurting alloy control. A **membrane** (an ion-selective divider) isolates the anode in its own compartment so the main bath's complexors are protected, while current still flows. Not every shop runs membranes, but if yours does, that compartment is part of your daily checks — never let it run dry or damaged.

★ REMEMBER

Three things make zinc-nickel chemistry different from plain zinc, and they all come from the same root — *it's an alloy with a sensitive sweet spot*: (1) **nickel held by amine complexors**, (2) **inert anodes with a separate metal feed**, and (3) **active alloy-composition control (%Ni)**. Internalize those three and the whole line makes sense.

CHAPTER 4

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

A zinc-nickel line is not one tank. It's a **sequence of tanks** a part travels through in a strict order — like a kitchen line or a car wash, where each station does one job and prepares the surface for the next. Parts ride down the line two ways:

- **Rack plating** — parts individually hung on a **rack** (a frame with hooks/clips that holds parts and carries current). Used for larger, critical, or cosmetic parts. Most automotive zinc-nickel rack work targets the tightest specs.
- **Barrel plating** — many small parts (fasteners, clips) tumbled together in a perforated rotating **barrel**. Efficient for high volumes. Because zinc-nickel is slow, barrel cycles run long.

> 💡 **TIP**
> You'll see different numbers for rack vs. barrel. In zinc-nickel, current density is **much lower** than acid zinc — roughly **1–4 A/dm² (10–40 ASF) on rack** and **0.3–1.2 A/dm² on barrel** — and because efficiency is low, **plate times are long.** Always check whether you're running rack or barrel and read the matching column on your shop-floor poster.

• THE WHOLE ZINC-NICKEL SEQUENCE: 9 STAGES

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Alkaline Soak Clean	Removes oil, grease, and shop soil	130–160°F	3–10 min
02	Rinse (Pre-Clean/Activation)	Washes off cleaner before activation	Ambient	30–60 s
03	Electroclean & Acid Activation	Final clean + dissolves oxide, "wakes up" steel	Amb–160°F	30 s–2 min
04	Rinse (Pre-Plate)	Guards the plating bath from acid & iron	Ambient	30–60 s
05	Zinc-Nickel Plating	Deposits the 12–16% Ni alloy	75–95°F	20–90 min
06	Rinse (Post-Plate)	Washes caustic off; decision point for bake	Ambient	30–60 s
07	Nitric/Acid Activation	Brightens; preps alloy surface for chromate	Ambient	5–30 s
08	Passivate (Trivalent)	Grows a chromate skin on the alloy	75–110°F	30–90 s
09	Seal / Topcoat + Dry	Seals for 720–1,000+ hr; gentle dry	Amb–cure	30–60 s + cure

The basic rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Activate → Passivate → Seal/Dry.**

> ★ **REMEMBER**
> Two things make this sequence different from a plain-zinc line: there's a **second activation (Station 07, often nitric)** right before passivation to clean and "wake up" the alloy surface, and there's a **seal/topcoat (Station 09)** because that's how zinc-nickel reaches its big salt-spray numbers. Everything else follows the familiar clean-rinse-activate-rinse-plate logic.

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the activation. A part that leaves the pickle still wet with acid (and iron) will contaminate the plating bath. What you do (or fail to do) at Stage 1 shows up as a defect — or a failed %Ni reading — at Stage 5. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the zinc-nickel tank is so precious.** Unlike the cleaner and the acids (dumped and remade regularly), a well-maintained zinc-nickel bath runs a long time — and it's expensive, full of nickel and complexors. Contaminants you let drag in don't just vanish: hexavalent chromium dragged back from the passivate, excess iron, or organics all degrade the alloy you're being paid to deposit. Every shortcut upstream charges interest at the most expensive tank on the line.

💡 **TIP**

Think of the zinc-nickel tank as the crown jewel of the line — and an *expensive* one, full of nickel and complexors. Everything *before* it exists to hand it a perfectly clean, bare, well-rinsed part. Everything *after* it (rinse, activate, passivate, seal) exists to protect and finish what the alloy created. The whole line orbits that one tank — and the alloy reading it produces.

A QUICK WORD ON "WHAT GOOD LOOKS LIKE"

At every handoff between tanks, there's a simple question you can learn to ask: *did the last station do its job?* A clean part sheets water in an unbroken film. A well-rinsed part reads low on the conductivity meter. A well-plated part comes out an even matte-to-bright gray and reads 12–16% Ni on the XRF. A well-activated alloy looks bright and uniform going into the passivate. A well-passivated part has even color with no blotches. You don't need a lab to catch most problems — you need to know what "good" looks like leaving each tank, and to stop when it doesn't.

★ **REMEMBER**

Each station has a visible or measurable "pass" sign: a water-break sheet, a conductivity number, an even deposit and an in-window %Ni, a bright activated surface, an even passivate color. Learn the pass sign for each tank, and you become the line's early-warning system.

CHAPTER 5

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here's the logic, link by link.

STAGE 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and chromate all need to touch *bare metal*. Oil and shop soil block them. **Plate over dirt and you've just plated dirt** — the alloy bonds to grease, not steel, and peels later. The free, foolproof check is the **water-break test**: water sheets off a clean part in an unbroken film; it beads up on a dirty one.

> ⚠ **HEADS-UP**
> The cleaner is **hot (130–160°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are before you need them.

STAGE 2 — RINSE (PRE-CLEAN/ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The soak cleaner film must come off before the part goes into the electroclean/activation tank, or it dilutes and contaminates what comes next. The rinse strips it.

> ★ **REMEMBER**
> *"Drag cleaner forward and you weaken the next tank; drag chemistry into the zinc-nickel bath and you poison it."* Rinses exist to keep each chemistry from invading the next tank.

STAGE 3 — ELECTROCLEAN & ACID ACTIVATION: MAKE IT SPOTLESS AND WAKE IT UP

Zinc-nickel demands an *immaculate*, oxide-free surface — its adhesion and alloy uniformity are unforgiving of leftover soil or oxide. So this station often does two things: an **electroclean** (cleaning with current applied, which scrubs the surface electrochemically) for a truly clean surface, then an **acid activation/pickle** that dissolves the invisible oxide and leaves chemically "fresh," active steel for the alloy to grip.

- **Under-prep** = soil/oxide remains = the alloy peels later (adhesion failure).
- **Over-pickle** = you etch and pit the part and load it with hydrogen.

> ★ **REMEMBER**
> **No clean, active surface — no adhesion, and no uniform alloy. Period.** Activation has to come right before plating, with only a rinse in between, because freshly activated bare metal grows a new oxide skin if it sits.

> ⚠ **HEADS-UP**
> Electrocleaning uses **current + hot caustic** (shock + burn hazards together). The acid dip uses **strong mineral acid** (HCl or H_2SO_4) that causes severe burns and corrosive fumes, and acid eating metal releases **flammable hydrogen gas.** Full PPE, good ventilation, and **always add acid to water, never water to acid.**

> ⚙ **TECHNICAL STUFF — HYDROGEN EMBRITTLEMENT STARTS HERE**
> During pickling and the low-efficiency zinc-nickel plating, atomic hydrogen diffuses into steel. In high-strength steel (**≥ 40 HRC / ≥ 180 ksi tensile**), trapped hydrogen can cause catastrophic delayed cracking under load — hours or days later. The fix is **baking** (375 ±25°F, ≥ 8 hours, started **within 4 hours** of plating; per ASTM B850). Use *inhibited* acids and minimum dip time on prone parts. A failed fastener or structural part can hurt people — always check the customer spec.

STAGE 4 — RINSE (PRE-PLATE): THE GUARD RINSE

This is the **guard rinse** protecting your most valuable tank. Acid dragged into the plating bath neutralizes precious free caustic, upsets the Zn:Ni ratio, and carries dissolved **iron** from the pickle into the alloy bath.

> ★ **REMEMBER**
>
> Alkaline zinc-nickel is somewhat more forgiving of iron than acid zinc (its filter catches precipitated iron) — but **don't get lazy.** Dragging acid into a caustic bath wastes caustic, disturbs the alloy-controlling ratio, and floods the filter. Keep this rinse tight: target **< 200 µS/cm**, double rinse strongly recommended.

> ⚙ **TECHNICAL STUFF**
>
> The recurring villain that justifies all those rinses is **drag-out** (and its mirror, **drag-in**). Drag-out is the film of solution clinging to a part as it leaves a tank; in the next tank it becomes drag-in — contamination. Operators measure rinse cleanliness with **conductivity**, in microsiemens per centimeter (µS/cm) — basically how many dissolved ions are in the water. *Higher conductivity = more dragged-in chemistry = a dirtier rinse.* Critical rinses on this line target **under 200 µS/cm.**

STAGE 5 — ZINC-NICKEL PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the alloy plating happen. The bath is high-pH caustic zincate with complexed nickel, running **75–95°F**, with **inert anodes** and the zinc/nickel fed separately while the **Zn:Ni ratio and temperature are actively managed** to hold the 12–16% Ni window.

> ⚠ **HEADS-UP**
>
> The plating bath is **hot caustic with nickel and amine complexors**, and gassing throws up a fine **caustic/metal mist** you must not breathe — local exhaust ventilation is required. The rectifier runs **high amperage** — shock and arc-flash risk. Because the bath is low-efficiency, lots of hydrogen is generated, raising embrittlement risk on parts above 40 HRC. LOTO before any rectifier work.

STAGE 6 — RINSE (POST-PLATE): GET THE CAUSTIC OFF + BAKE DECISION

The plating bath is loaded with **caustic and nickel.** Drag it forward and it neutralizes the acid activation and contaminates the passivate. Many shops make the first post-plate rinse a **drag-out recovery (dead) rinse** — a still tank whose nickel-bearing water is pumped back to top off the plating bath, saving expensive chemistry and cutting nickel waste.

> ★ **REMEMBER**
>
> **Caustic and nickel dragged into the finishing tanks are wasteful and damaging.** This rinse is the wall between the high-pH plating bath and the low-pH finishing tanks. It's also the decision point for the hydrogen-embrittlement bake on hardened steel.

STAGE 7 — NITRIC/ACID ACTIVATION: WAKE UP THE ALLOY FOR CHROMATE

Fresh zinc-nickel comes out of a high-pH bath; its surface needs to be **brightened and chemically activated** for the passivate to react evenly. A quick **dilute nitric (or other acid) dip** removes the dull caustic film and any light oxidation, leaving a clean alloy surface so the chromate builds a uniform film and the right color.

> ★ **REMEMBER**
>
> This little dip is easy to underrate. **A poor activation here = a blotchy, weak, wrong-colored passivate.** It's the bridge between plating and passivating.

HEADS-UP

Nitric acid is a **strong oxidizer and a severe burn/fume hazard.** It can react dangerously with organics and metals and releases irritating brown fumes. Ventilation required, full acid PPE, and never mix it with other tank chemistries. Acid-to-water always.

STAGE 8 — PASSIVATE: LOCK IN THE FINISH

The passivate (a trivalent chromate conversion coating) grows a thin protective skin on the alloy that dramatically slows **white rust** and can add color — clear/blue or black are most common on zinc-nickel.

HEADS-UP

Even trivalent passivate is **acidic (pH ~1.5–4) and causes burns.** If any **hexavalent chromium (Cr^{6+})** is in use, it's a **confirmed human carcinogen** (OSHA PEL 5 μg/m³) requiring supplied-air respirators and medical surveillance under OSHA 29 CFR 1910.1026. Know which chemistry your shop runs — some legacy ZnNi blacks were hexavalent.

STAGE 9 — SEAL/TOPCOAT, FINAL RINSE & DRY: REACH FOR 1,000 HOURS

A **seal or topcoat** (often applied warm, then cured) is what pushes zinc-nickel to its premium salt-spray numbers and can add lubricity (controlled torque-tension on fasteners). A final rinse and a *gentle* dry/cure finish the part without cracking the fresh films.

REMEMBER

The whole nine-stage sequence is one continuous logic chain: *clean it so the alloy can stick → rinse so the cleaner doesn't ruin activation → electroclean and activate so the metal is immaculate and bare → rinse so acid and iron don't poison the precious alloy bath → plate the 12–16% Ni alloy → rinse so caustic and nickel don't kill the finishing tanks → activate so the chromate takes evenly → passivate to protect the alloy → seal and gently dry so it ships at 1,000 hours.* Every step earns its place. Skip one, and the chemistry of the next step tells on you.

TIP

When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. A peeling deposit? Look upstream at cleaning and activation. A blotchy passivate? Look at the post-plate rinse and the nitric activation. A failing %Ni reading? Look at temperature and the Zn:Ni ratio. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 6

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

A zinc-nickel line works with hot caustic, nickel salts and amine complexors, strong mineral acids (including nitric), electrical current at high amperage, chemical mists, and chromium chemistry. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

★ **REMEMBER**

There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes cannot.

THE MASTER PPE SET (WEAR AT ALL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — *over* the goggles for any work involving acids, hot caustic, the plating bath, the nitric dip, or passivate.
- **Chemical-resistant gloves** — elbow-length, rated for caustics and acids (nitrile, neoprene, or butyl depending on the chemistry; **butyl/neoprene for nitric**). Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

⚠ **HEADS-UP**

Respiratory protection is required when ventilation is inadequate or when working over fuming tanks (electroclean/caustic mist, acid pickle, the nitric dip, plating-bath mist, passivate). **Nickel-bearing mist is a respiratory sensitizer and a regulated metal** — never breathe it. Use the cartridge or supplied-air respirator your shop specifies. If you can smell acid or ammonia/amine fumes strongly, ventilation isn't keeping up — stop and report it.

💡 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

STATION-BY-STATION HAZARD SUMMARY

STAGE	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak Clean	**Hot alkaline (caustic)**, 130–160°F	Chemical burns; hot splashes
02 Rinse	Residual alkaline cleaner	Mild irritant; wet/slip hazard
03 Electroclean & Acid Activation	**Hot caustic + current**; **strong mineral acid**; HCl fumes; **H_2 gas**	Burns, shock, corrosive fumes, fire risk
04 Rinse	Dilute acid residue	Mild irritant; slip hazard
05 Zinc-Nickel Plating	**Caustic + nickel + amine mist**; **high-amperage rectifier**	Burns; respiratory sensitizer; shock/arc-flash
06 Rinse	Dilute caustic + nickel residue	Irritant; nickel exposure; slip hazard
07 Nitric/Acid Activation	**Nitric acid**; oxidizer; brown fumes	Severe burns; corrosive/oxidizing fumes
08 Passivate	**Acidic chromate** (Cr(III) pH 1.5–4; or Cr(VI) where used)	Burns; Cr(VI) is a **carcinogen**
09 Seal/Dry	Hot seal/oven; residual chemistry	Burns from hot dryer; slip hazard

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially heavy metals like **nickel** and chromium. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

⚠ **HEADS-UP — NICKEL SENSITIZATION**

Nickel is a well-known **skin and respiratory sensitizer** — repeated exposure can cause an allergy that, once developed, can be permanent. Keep nickel solutions off your skin, out of cuts, and out of your breathing air. This is a reason zinc-nickel demands *more* PPE discipline than plain zinc, not less.

WHY ZNNI PPE IS A NOTCH STRICTER

If you've worked a plain-zinc line, the gear here will feel familiar — goggles, shield, gloves, apron, boots. What's different is the *respiratory* discipline and the *no-skin-contact* rule, both driven by the nickel and the amine complexors. On plain zinc you mostly guard against burns. On zinc-nickel you guard against burns *and* against a metal that can sensitize you for life if you're careless with the mist. Treat the respirator and the gloves as non-negotiable, not as "extra" gear for fussy days.

💡 **TIP**

Keep a spare pair of intact gloves at your station. The moment you spot a pinhole, a thin spot, or a chemical that has wicked inside, swap them — don't "finish the rack first." A compromised glove on a nickel/caustic line is worse than no glove, because it traps chemistry against your skin.

★ **REMEMBER**

The master set goes on every time, at every station: **goggles, face shield, chemical gloves, apron, boots** — plus the **respirator** wherever mist or fumes are present. Nickel and amines mean you protect your *lungs and skin* as carefully as your eyes here.

CHAPTER 7

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, and the **spill kit** and **SDS binder** (the official hazard documents for every chemical on site).

⚠ **HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Safety shower; flush **at least 15 minutes**. Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS.
Chemical in eyes	Eyewash; hold eyelids open and flush **at least 15 minutes**, rolling your eyes. Don't rub. Don't stop early. Get medical attention immediately — eye exposures are always urgent. Bring the SDS.
Inhaled fumes/mist (esp. nickel or nitric), feel ill	Fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	**Do not touch the person** if they may still be in contact with current. Cut power at the emergency stop/disconnect first, then call for help and render aid.
Chemical spill	Alert others; keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

LOTO is mandatory for *all* rectifier and electrical maintenance. It means physically locking the power off and tagging it so no one can switch it back on while someone is working on it. "I'll just be a second" is how people get electrocuted.

⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid (HCl, H_2SO_4, **and especially nitric**), **always add ACID to WATER — never water to acid.** Adding water to concentrated acid releases heat so violently it can boil and erupt acid into your face. The memory aid: *"Do as you oughta — add acid to water."* Pour slowly, down the side, acid going *into* a larger volume of water.

⚠ **HEADS-UP — DON'T CROSS-MIX TANK CHEMISTRIES**

Nitric acid is a strong oxidizer; mixing it with the caustic plating bath, with cyanide (if any legacy chemistry is on site), or with organics can react violently or release toxic gas. Keep every chemistry in its own labeled container and tank. When in doubt, ask before you pour.

☀ **FUN FACT**

The reason acid-to-water matters is heat capacity: water absorbs the large heat of dilution far better when the acid is the small addition to a big pool of water. Get it backwards and the tiny bit of water flash-boils and spits acid. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 8

THE SAFETY PHILOSOPHY OF A ZINC-NICKEL LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even rinse tanks carry residual chemistry, and rinses here carry nickel. Even “just water” can be slippery enough to put you on the floor. Respect comes first; familiarity is where complacency — and accidents — begin.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don’t fling racks, splash, or rush handoffs. Most chemical exposures aren’t dramatic spills — they’re splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses and better alloy control).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

The things that make good parts are the same things that can hurt you. The caustic that dissolves zinc burns your skin. The nickel that delivers corrosion resistance is a sensitizer. The current that deposits the alloy can arc-flash. The chromate that protects the part is toxic. You manage the work and the hazard at once, every time.

4. THE CHEMISTRY IS DEMANDING, SO BE DISCIPLINED.

Zinc-nickel has more knobs than plain zinc — two metals, a ratio, complexors, alloy composition. The same discipline that keeps the bath and the alloy in spec — careful rinsing, no shortcuts, controlling drag-in — is the discipline that keeps *you* healthy. Good housekeeping is a safety practice, not just a quality practice.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer while learning. The unsafe operator isn’t the one who asks questions — it’s the one who guesses. If a fume seems strong, a glove looks worn, a %Ni reading looks wrong, or a procedure feels off, *stop and check.*

• THE FOUR HAZARD FAMILIES TO INTERNALIZE

1. **Caustic & acids (corrosives).** The hot caustic cleaner and plating bath (high pH) and the mineral acids — especially **nitric** at Station 07 — cause burns on contact and damage eyes instantly. *Defense:* goggles, face shield, gloves, apron, the 15-minute flush rule, and acid-to-water always.
2. **Nickel & amine complexors (toxic/sensitizing).** Nickel is a skin/respiratory sensitizer and a regulated metal; some complexors are exposure and waste concerns. *Defense:* keep solutions and mist off skin and out of lungs, wear the specified PPE and respirator, never eat/drink on the line, wash thoroughly.
3. **Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. *Defense:* LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, clear/dry walkways.
4. **Chromate / passivation chemistry.** Trivalent Cr(III) is an acidic irritant; hexavalent Cr(VI), if present anywhere, is a carcinogen with strict regulatory controls. *Defense:* know which your shop uses, wear the specified PPE, control chromate mist, follow OSHA requirements to the letter for any Cr(VI).

★ **REMEMBER**

Caustic and acids, nickel and complexors, electrical and physical, chromate chemistry. Four families. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

★ **REMEMBER**

You've now got the whole foundation: what plating is, what makes zinc-nickel special, what an alloy deposit is and why the 12–16% Ni sweet spot rules everything, how the nine-stage line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

☀ **FUN FACT**

The best operators aren't the ones who never have a problem — they're the ones who catch a problem early, while it's still a cheap fix. A safety mindset and a quality mindset are the same mindset: pay attention, move deliberately, and trust what the tank in front of you is telling you.

PART TWO — THE PREP LINE (STATIONS 01–04)

STATION 01

ALKALINE SOAK CLEAN

“PLATE OVER DIRT AND YOU’VE JUST PLATED DIRT.”

★ **REMEMBER — THE PREP-LINE PROMISE**

By the time a part reaches the zinc-nickel tank, most of whether it’ll turn out beautiful or scrap has already been decided. The four prep stations — Clean, Rinse, Activate, Rinse — are not the “boring stuff before the real plating.” As the old-timers say: *“The plating bath gets the credit, but the pretreatment does the work.”* Zinc-nickel is even less forgiving than plain zinc: a so-so clean shows up as poor adhesion AND uneven alloy.

Welcome to the very first tank. Your job is simple to say and important to nail: **get the part perfectly clean.** Not “looks clean.” *Chemically* clean — no oil, no grease, no fingerprints, no shop dust. Do this right and you make every job after you easier. Cut a corner here, and it comes back as peeling, hazing, or spotty plating that nobody can fix later.

• WHAT YOU’RE DOING & WHY IT MATTERS

Steel parts arrive carrying three layers of stuff: **particulate** (loose dust, metal fines, grit), **oil and grease** (the slippery, often invisible film from stamping oils, cutting fluids, rust-preventives, and hands), and **oxide scale** (thin rust bonded to the steel — removed at Station 03, not here).

Soak Clean removes the first two. The cleaner is a hot, strongly alkaline solution that works several ways at once: **heat** thins the oils so they let go; **surfactants** (the soap-like ingredients) surround oil droplets and lift them off, like dish soap on a greasy pan; and **builders/alkalinity** chemically break down (saponify) certain oils so water can carry them away.

💡 **TIP — THE WATER BREAK TEST (FREE, FAST, NEVER LIES)**

A truly clean steel surface loves water. Pull a clean part and the water clings and **sheets off in one continuous, unbroken film.** A dirty surface repels water, so it **beads up** into droplets. **Continuous sheet = PASS; beading/broken film = FAIL** (re-clean). Costs nothing, takes two seconds. Use it on every rack, every time.

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WATER-BREAK TEST  ( pull part, watch the water )

FAIL  (oil remains)           PASS  (clean steel)
.  o   o  .   o                ==================
 o   .  o   .                  ~~~~~~~~~~~~~~~~~~
o  .  o   o  .                 continuous unbroken
water BEADS up                water SHEETS off
-> re-clean                   -> go to Station 02
```

⚙ **TECHNICAL STUFF**

“Water break” reflects the **surface energy** of the metal. Clean steel has high surface energy, so water spreads (low contact angle). Even a one-molecule film of oil drops the surface energy, so water pulls into beads (high contact angle). That’s why the test catches contamination too thin for your eye to see.

• OPERATOR ESSENTIALS (memorize the ranges)

PARAMETER	RANGE	WHAT TO WATCH
Temperature	130–160°F (54–71°C)	**Do not exceed 180°F**
Time	3–10 min	Heavy oil: 8–10 min · Light: 3–5 min
Concentration	4–8 oz/gal (30–60 g/L)	Titrate (test) weekly at minimum
pH	11.0–13.5	Depends on the cleaner formulation
Agitation	Air or mechanical, continuous	Still soak for tubular parts
Filtration	Surface skimmer recommended	Removes floating oil
Water Break Test	Pass = continuous sheet	The only reliable field test

Plain-language definitions: **Concentration** = how much cleaner is dissolved in the water; too weak won't cut oil, too strong wastes money. **Titrate** = a simple test (QC/lead can show you) that measures actual concentration. **Agitation** = movement that keeps fresh cleaner against the surface. **Surface skimmer** = pulls floating oil off the top so it doesn't redeposit.

⚠ **HEADS-UP — HOT ALKALINE SOLUTION**
This tank is **hot AND caustic.** Splashes cause **chemical burns on contact**, and the steam/mist irritates airways. **PPE every time:** splash goggles, face shield (charging/dumping), elbow-length chemical gloves, apron, boots. First aid: **skin → flush 15 minutes; eyes → eyewash 15 minutes** and get help. Never lean over the tank without eye protection.

• STEP-BY-STEP PROCEDURE

1. **Check the tank before you start.** Temperature **130–160°F** (never above 180°F); concentration in range (check the titration log); skimmer running, surface reasonably oil-free.
2. **Inspect the load.** Heavy rust-preventive oil → 8–10 min. Light stamping oil → 3–5 min.
3. **Rack or load the parts** so solution reaches every surface. Avoid nesting where air can be trapped (trapped air = a dry spot = a dirty spot). Position tubular/blind-hole parts so the bore floods and drains.
4. **Immerse fully** and start your timer. Confirm agitation is working (or a still soak for tubulars per shop spec).
5. **Soak the full time.** Don't rush — the clock is doing chemistry you can't see.
6. **Withdraw slowly,** letting solution drain back (saves chemical, cuts drag-out).
7. **Run the water break test.** Continuous sheet → Station 2. Beading → back in the cleaner.
8. **Log it** if your shop logs cleaning, and move the rack promptly so the clean surface doesn't flash-rust.

• TEACH-BACK CHECKLIST

A trainee is signed off on Station 01 only when they can, unprompted:

- Name the two kinds of contamination the soak clean removes (particulate and oil/grease — NOT oxide; that's Station 3).
- State the temperature range and hard ceiling (**130–160°F; never over 180°F**).
- State the time for heavy vs. light oil (8–10 min vs. 3–5 min).
- Correctly perform a water-break test and read it honestly (continuous sheet = pass).
- Walk through what to do if the water-break fails (re-clean it — never send it forward).
- Name three pieces of required PPE.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Skipping the water-break test because the part "looks fine."** Oil that ruins plating is invisible.
2. **Running the tank too cold.** Cold cleaner can't cut oil. If the water-break keeps failing, check temperature first.
3. **Letting concentration drift low.** If nobody's titrating, you're slowly plating dirt without knowing it.
4. **Pulling parts through the floating oil layer.** A dead skimmer redeposits the oil you just removed.
5. **Nesting parts so tight that air pockets form.** Wherever solution can't reach stays dirty.
6. **Going too hot to "clean faster."** Above 180°F you flash off solution, etch parts, and make more mist hazard.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; bring temp to range
Oily film after cleaning	Floating oil redepositing	Clean/run skimmer; dump tank if heavily contaminated
White residue on parts	Hard-water salts; cleaner residue	Check rinse-water quality (use DI)
Etching/discoloration	Cleaner too aggressive or too hot	Reduce temp; consider milder formulation
Spotty adhesion downstream	Uneven cleaning; trapped air	Reposition on rack; add agitation

SIGN-OFF — STATION 01

Trainee ____________ Trainer ____________ Date ____________

"I confirm the parts in this load passed the water break test (continuous sheet, no beading), the soak clean was at 130–160°F and in concentration, and I wore required PPE."

FUN FACT

The water-break test is older than most plating chemistry on your line — metal finishers were "reading the water" off cleaned parts long before anyone could measure surface energy with an instrument. It's free, instant, and still one of the most trusted checks in the trade.

STATION 02

RINSE (PRE-CLEAN/ACTIVATION)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

You just got the part beautifully clean — but it’s now coated in a thin film of the cleaner itself. This station’s whole job is to **wash that cleaner film off** before the part goes into the electroclean/activation tank. It sounds like a “nothing” step. It isn’t: a weak rinse here quietly sabotages the next tank, and you won’t see the damage until parts come out wrong.

• WHAT YOU’RE DOING & WHY IT MATTERS

The soak cleaner is **alkaline** (high pH). Carrying that film forward dilutes and contaminates the electroclean and, downstream, drags toward the acid activation where alkaline and acid neutralize and cancel each other out. So a poor rinse here means: **you waste downstream chemistry**, **you weaken activation**, and **you reduce activation quality** — which shows up later as poor adhesion (alloy that peels).

This invisible carry-over is called **drag-out** (leaving the tank on the part) or **drag-in** (arriving in the next tank). The rinse knocks it down by **dilution** — clean water surrounds the part, the cleaner film dissolves into the rinse water, and the constant flow of fresh water flushes it away.

> 💡 **TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY**
> You can’t see dissolved cleaner, so we measure it with **conductivity** — how well the water conducts electricity. Pure water barely conducts; dissolved chemicals make it conduct *more*. So **higher conductivity = more cleaner left in the rinse = dirtier rinse.** Measured in **microsiemens per centimeter (µS/cm)** — just “the contamination number.” Lower is cleaner.

> ★ **REMEMBER — THE RULE CARD NUMBER: < 500 MS/CM**
> Keep this rinse **below 500 µS/cm** (aim **under 300**). Above 500, you’re dragging alkaline cleaner forward — weakening the next tanks. Check it daily.

• THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* just blast fresh water through a single tank and dump it — it works but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: parts move forward; water flows backward. Fresh water enters the *last* (cleanest) tank and overflows backward into the dirtier tank, so the part gets progressively cleaner rinses, and the cleanest water it sees is the last thing to touch it.

```
COUNTER-FLOW RINSE  ( parts ->   water <- )

parts in            RINSE 1          RINSE 2          to ELECTROCLEAN/
(dirtiest) -------> [dirtier] -----> [cleaner] -----> ACTIVATION
                       ^                 |
                       |  overflow back  |
                       +---- WATER <------+ fresh DI in
cleanest water meets the cleanest parts -> least drag-in, least waste
```

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	30–60 sec with agitation	Longer for blind holes/tubing
Water Source	Municipal or DI	**DI preferred**
Conductivity	< 500 µS/cm (target < 300)	Monitor daily
Flow Rate	2–4 gal/min	Continuous overflow or counter-current
Tank Turnover	4–6 / hour	Prevents stagnation
pH	7–10	Elevated pH = cleaner drag-in

Plain-language definitions: **Ambient** = room temperature, no heater. **DI water** = deionized water with minerals removed (low conductivity, rinses better, no spots). **Continuous overflow** = fresh water in, dirty spills out the top. **Tank turnover** = how many times per hour the whole tank volume is replaced.

⚠ **HEADS-UP — RESIDUAL ALKALINE CLEANER**
The rinse water carries dragged-in cleaner and starts mildly caustic. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and slips are the #1 injury here.

• STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** Near or above 500 µS/cm → tell your lead; the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (continuous overflow, or counter-current cascade active).
3. **Transfer the rack promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Immerse fully with agitation, 30–60 seconds.** For blind holes and tubing, go longer and move the part to flush trapped solution.
5. **Lift and drain** over the rinse tank for a few seconds.
6. **Move promptly to the next tank** — don't let the rinsed part sit and flash-rust.

• TEACH-BACK CHECKLIST

- Explain why carrying cleaner forward causes problems (dilutes/neutralizes the next tanks; weakens activation).
- Name the term for chemistry clinging to a part from the previous tank (drag-out / drag-in).
- Explain what conductivity tells us and which direction is "clean" (lower = cleaner).
- State the conductivity limit and target (**< 500 µS/cm; target < 300**).
- State how long to rinse and why longer for tubing (30–60 sec; tubing traps solution).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Heavy drag-out; low flow	More drain time off cleaner; more rinse flow; add counter-current stage
Soapy residue on parts	Inadequate rinse time	Extend dwell; add agitation
Next tank going off-balance	Cleaner carry-over	Improve this rinse; add a second stage

SIGN-OFF — STATION 02

Trainee ______________ Trainer ______________ Conductivity __________ µS/cm

"I confirm the pre-clean/activation rinse conductivity was within limit (< 500 µS/cm), water flow was running, and parts were rinsed the full time with PPE worn."

STATION 03

ELECTROCLEAN & ACID ACTIVATION

"NO CLEAN, ACTIVE SURFACE — NO ADHESION. PERIOD."

This is the station that makes the alloy *stick*. Zinc-nickel is unforgiving: its adhesion and its alloy uniformity both depend on a surface that's not just clean but *immaculate* and oxide-free. So this station usually does two jobs in sequence — an **electroclean** (cleaning with current applied) to get a truly spotless surface, then an **acid activation/pickle** to strip the invisible oxide and leave fresh, reactive steel. This is also the station with the **most safety weight** in your section — hot caustic, current, strong acid, fumes, and the start of hydrogen embrittlement.

PART A — ELECTROCLEAN: CLEANING WITH CURRENT

A regular soak clean lifts most soil; an electroclean finishes the job by running **current** through the part while it sits in a hot alkaline cleaner. The current generates tiny gas bubbles right at the surface that scrub off the last clinging film — a microscopic power-wash. Most lines run **anodic** electrocleaning (part positive) for steel, because it avoids driving hydrogen *into* the part; some use a brief cathodic or periodic-reverse step per the supplier's process.

> ⚙ **TECHNICAL STUFF — ANODIC VS. CATHODIC CLEAN**
>
> In **cathodic** cleaning the part is negative and hydrogen forms *on the part* — great scrubbing but it loads hydrogen into the steel (bad for embrittlement). In **anodic** cleaning the part is positive and oxygen forms on the part instead — less embrittlement risk, and it helps remove smut. For high-strength steel, anodic (or periodic-reverse ending anodic) is the safer choice. Follow your shop's process exactly; this isn't an operator guess.

PART B — ACID ACTIVATION: WAKE UP THE METAL

Even a spotless steel surface grows an invisible **oxide** film in minutes in air. The alloy can't bond to oxide — only to bare, chemically "awake" steel. The acid dip dissolves that oxide and leaves a fresh, reactive surface. You'll see tiny bubbles — that's **hydrogen gas (H_2)**, normal, and a clue to the hidden hazard below.

> ⚙ **TECHNICAL STUFF**
>
> The acid (say HCl) reacts with iron oxide to dissolve it (iron oxide + acid → soluble iron salt + water); once the oxide is gone the acid attacks the iron itself ($Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$). That last reaction is why you must *time the dip* — a little attack cleans and activates; too much etches the part and floods it with hydrogen.

> ★ **REMEMBER — THE RULE CARD: TIME EVERYTHING**
>
> Total processing here is short (roughly 30 sec–2 min across electroclean + pickle, per shop process). **Over-process** = over-etch (roughness, pitting) AND excess hydrogen in the steel. **Under-process** = soil/oxide remains and the alloy won't stick. Find the right times and *time them.*

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Electroclean temp	120–160°F (49–71°C)	Hot alkaline + current
Electroclean current/time	Per shop process, 30–90 sec	**Anodic preferred** for high-strength steel
Pickle temperature	Ambient–110°F	Ambient typical; warm only for heavy scale
Pickle time	15–60 sec	15–30 clean steel · 30–60 heavy oxide
Acid — Option A	HCl 10–50% v/v	Common; fumes — **ventilation required**
Acid — Option B	H_2SO_4 5–25% v/v	Less fuming; slightly slower
Agitation (pickle)	Manual rack movement only	**No air agitation** — it creates acid mist
Iron Content (pickle)	Dump when > 50 g/L Fe	Spent acid loses bite; iron can drag forward
Inhibitor	Per TDS (use on > 40 HRC)	Reduces base-metal attack & hydrogen absorption

Plain-language definitions: **HCl** = hydrochloric (muriatic) acid; **H_2SO_4** = sulfuric acid — both strong mineral acids. **% v/v** = "percent by volume" (50% v/v HCl = half acid, half water). **Inhibitor** = an additive that lets the acid keep dissolving *oxide* while slowing its attack on the *base steel*, protecting the part and cutting hydrogen absorption. **Iron content (Fe)** = dissolved iron that builds up; over 50 g/L the acid is "spent."

> ⚠ **HEADS-UP — HOT CAUSTIC + CURRENT + STRONG ACID (READ THIS TWICE)**
>
> The electroclean is **hot caustic with electrical current** — burn *and* shock hazards together; never reach in energized. **HCl fumes at room temperature** — ventilation required. **H_2SO_4 causes severe dehydration burns** and reacts violently with water. **Hydrogen gas (H_2) from pickling is flammable** — no open flames or sparks near the tank. **PPE:** goggles + face shield, acid-resistant gloves, apron, boots, **respirator if ventilation is inadequate.** Skin/eyes: flush 15 min. **Spills:** neutralize acid with soda ash; **never pour water onto concentrated acid.** Always add acid to water. **No air agitation** in the pickle — bubbling air throws acid mist into your breathing space; move the rack by hand.

• THE BIG ONE: HYDROGEN EMBRITTLEMENT

> ⚠ **HEADS-UP — HYDROGEN EMBRITTLEMENT (THE MOST IMPORTANT CONCEPT IN YOUR SECTION)**
>
> When acid attacks steel — and when the low-efficiency zinc-nickel bath plates — hydrogen is released. Most bubbles away, but some sneaks *into* the steel as single hydrogen atoms wedged between iron atoms. On ordinary mild steel this is usually harmless. On **high-strength/hardened steel (≥ 40 HRC / ≥ 180 ksi UTS)** it makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later. On a critical fastener or structural part, that's a real safety failure.
>
> **What you do about it:** Use the **minimum effective acid concentration and dip time** — don't over-dip "to be safe" (the opposite of safe here). **Use an inhibitor on substrates over 40 HRC.** Prefer **anodic** electrocleaning. If a job is flagged high-strength or embrittlement-sensitive, **stop and check with your lead** — and remember that **zinc-nickel is a low-efficiency, high-hydrogen process, so the post-plate bake (Station 06) is mandatory** on such parts.
>
> *Terms:* **HRC** = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of psi of ultimate tensile strength. You don't have to calculate these — recognize the words on a traveler and treat them as a "slow down and check" flag.

• STEP-BY-STEP PROCEDURE

1. **Check the job for hardness flags first.** High-strength, hardened, ≥ 40 HRC, or ≥ 180 ksi → pause, confirm process, anodic clean, and inhibitor.
2. **Confirm ventilation is running** and PPE (including face shield) is on.
3. **Electroclean** per shop process (anodic typical), then rinse if your line has an interstage rinse.
4. **Verify acid strength and dip time:** 15–30 sec for clean mild steel; 30–60 sec for cast iron/heavy oxide.
5. **Immerse fully, start your timer, agitate by hand only** — never air.
6. **Watch the surface.** Light bubbling is normal; the dark oxide should lighten/disappear. Not clearing? Acid may be dilute or spent — check the bath; don't just leave a hard part longer.
7. **Pull at the timed mark.** Don't "round up." Over-dipping etches and embrittles.
8. **Drain briefly and move promptly** to the pre-plate rinse (Station 4) — activated steel re-oxidizes fast.
9. **Watch the iron level** over time; flag for dump/recharge at **> 50 g/L Fe.**

• TEACH-BACK CHECKLIST

- Explain the two jobs here (electroclean for a spotless surface; acid pickle to strip oxide and activate).
- Explain anodic vs. cathodic clean and which is safer for hard steel (anodic — avoids driving hydrogen in).
- Explain over-dip vs. under-dip (over: etching + embrittlement; under: oxide remains, alloy peels).
- State pickle times for clean steel vs. heavy oxide (15–30 vs. 30–60 sec).
- Explain why there's NO air agitation in the pickle (acid mist).
- Explain hydrogen embrittlement and which parts are at risk (≥ 40 HRC / ≥ 180 ksi), and that ZnNi's low efficiency makes the post-plate bake mandatory on those parts.
- Recite the acid-dilution rule (always add acid to water).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains	Acid too dilute; time too short	Increase concentration; extend time *(check embrittlement risk first)*
Pitting / etching	Acid too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure downstream	Incomplete clean/activation; residual oxide or soil	Re-check electroclean and acid strength/time; check alkaline drag-in from Station 2
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective; check exhaust
Dark smut on surface	High dissolved iron; copper contamination	Dump/recharge acid; investigate copper source
H-embrittlement failures	Over-long pickle; cathodic clean on hard steel; no inhibitor; skipped bake	Shorten pickle; switch to anodic; add inhibitor; bake per ASTM B850

SIGN-OFF — STATION 03

Trainee ____________ Trainer ____________ Pickle _______ sec · Inhibitor? Y/N · Clean: Anodic/Cathodic

"I confirm I checked the job for hardness flags, ran the electroclean (anodic where required) and acid pickle at timed intervals with manual agitation only, ventilation and PPE in place, and used inhibitor where required."

STATION 04

RINSE (PRE-PLATE)

THE GUARD RINSE — THIS RINSE PROTECTS THE MOST EXPENSIVE TANK ON THE LINE

You're at the last station before the zinc-nickel bath — the tank full of nickel and complexors that makes the company's premium money. Your job: **wash the acid (and the dissolved iron it carries) off the part** so it doesn't contaminate that expensive, sensitive plating bath. Think of this rinse as the **guard at the door** of the most valuable room in the building.

• WHAT YOU'RE DOING & WHY IT MATTERS

Coming out of the pickle, the part is wet with acid that now contains **dissolved iron**. Carry that into the plating bath and you cause three problems:

1. **You neutralize precious free caustic.** The alkaline zinc-nickel bath depends on its NaOH level; dragging in acid eats caustic and upsets the **Zn:Ni ratio** that controls your alloy.
2. **You stress the complexor/brightener system.** Acid drag-in disturbs the delicate chemistry that holds nickel in solution and controls codeposition — meaning alloy drift and appearance problems.
3. **You add iron.** Iron drives haze and dark deposits and burdens the filter.

★ **REMEMBER — IRON IN ZINC-NICKEL**

Unlike acid zinc (where iron stays dissolved forever), an alkaline zinc-nickel bath can **precipitate iron and catch it in the filter** — so it's somewhat more forgiving. But don't relax: heavy iron drag-in still floods the filter, wastes caustic, and disturbs the alloy balance. **This pre-plate rinse is your primary defense.** Hold it tighter than any rinse before it.

CONTAMINANT	SOURCE	EFFECT ON THE ZNNI BATH	PREVENTION
Iron (Fe)	Spent pickle	Haze, dark deposits; filter burden	Refresh pickle; double rinse; hold conductivity tight
Free acid	HCl/H_2SO_4 drag-out	Neutralizes caustic; upsets Zn:Ni ratio	Drain well; tighter rinse
Acid + chemistry load	Excess drag-in	Stresses complexor/brightener balance → alloy drift	Add a second rinse stage

💡 **TIP — CONDUCTIVITY, TIGHTER THIS TIME: < 200 MS/CM**

Same "contamination number" as Station 2, but here the limit is **under 200 µS/cm, target under 100.** Tighter, because acid and iron drag-in directly disturb the alloy chemistry and shorten bath life. Check daily; don't let it creep.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	30–60 sec with agitation	**Double rinse recommended**
Water Source	DI preferred	Minimizes contamination to plating bath
Conductivity	< 200 µS/cm (target < 100)	Critical for ZnNi bath/alloy stability
pH	4.0–8.0	Below 4 = excessive acid drag-in
Flow Rate	Continuous overflow, 3–5 gal/min	Counter-current ideal
Drain Time	15–20 sec minimum	Prevents diluting the plating bath

Plain-language notes: **Double rinse** = two rinse tanks in a row; the first knocks off most acid, the second polishes to the tight target. **Drain time** matters most here — a solid 15–20 seconds means less contamination into the money tank. **pH below 4** warns you're dragging too much acid.

⚠ **HEADS-UP — DILUTE ACID RESIDUE**
The water here carries dragged-in acid — diluted but irritating. Wear **goggles, chemical gloves, and non-slip footwear.** Floor is wet; watch your footing.

★ **REMEMBER — THE 30-SECOND CLOCK**
Move into the plating bath within ~30 seconds of leaving the rinse. A freshly activated, rinsed surface oxidizes in air almost immediately, and an oxidized surface plates poorly and unevenly — undoing Station 3's work. Keep the part wet; speed matters here.

• STEP-BY-STEP PROCEDURE

1. **Check conductivity** at shift start. Above 200 µS/cm → more flow or a refresh before you plate through it.
2. **Confirm water flow** (continuous overflow 3–5 gal/min; counter-current cascade if equipped).
3. **Transfer promptly from the pickle** — don't let the activated, acid-wet part re-oxidize.
4. **Immerse fully with agitation, 30–60 seconds.** Give both tanks their time if double-rinsing.
5. **Drain a full 15–20 seconds** over the rinse. Non-negotiable here.
6. **Move into the plating bath within ~30 seconds** of leaving the rinse.

• TEACH-BACK CHECKLIST

- Name the three problems acid/iron drag-in causes in the ZnNi bath (neutralizes caustic; upsets Zn:Ni ratio & complexors; adds iron).
- Explain how iron behaves in alkaline ZnNi vs. acid zinc (precipitates/filters here — more forgiving — but still keep it tight).
- State the conductivity limit here vs. Station 2 (< 200, target < 100, vs. < 500).
- State drain time and why (15–20 sec; less drag-in into the expensive bath).
- State how fast the part must reach the plating tank (within ~30 sec, before it re-oxidizes).

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Conductivity high	Heavy acid drag-out; low flow	Increase drain time off pickle; increase flow
Bath caustic/ratio drifting	Acid drag-in from pickle	Tighten conductivity; add a double rinse
Iron/filter load climbing	Dissolved Fe dragging in from spent pickle	Dump/refresh pickle; improve rinse; add a stage
Parts oxidizing before plating	Transfer too slow; part drying out	Speed up — into plating within ~30 sec

SIGN-OFF — STATION 04

Trainee ______________ Trainer ______________ Conductivity ___________ µS/cm

"I confirm the pre-plate rinse conductivity was within limit (< 200 µS/cm), parts were rinsed and drained the full time, and moved to plating within ~30 seconds with PPE worn."

★ **REMEMBER — THE PREP-LINE PROMISE**
Four stations, one mission: hand the alloy bath a surface that is **clean, contaminant-free, oxide-free, and still wet.** Do these four right, every rack — and a uniform, well-bonded 12–16% Ni alloy will reward you.

PART THREE — THE MAIN LINE (STATIONS 05–09)
STATION 05

ZINC-NICKEL PLATING

WHERE BARE STEEL BECOMES CORROSION-ARMORED STEEL

Take a breath. Up to now, every station has had one job: get the steel ready. None of those tanks put any metal on the part. **Station 05 is where the zinc-nickel alloy coating is actually created.** Everything before it was preparation; everything after it is protection and finishing. If you only deeply understand one station on the whole line, make it this one.

The big idea in one sentence: **we use electricity to pull dissolved zinc *and* nickel out of a caustic liquid and lock them, atom by atom, onto your steel part as a 12–16% nickel alloy.** Meet the cast:

- **The cathode** — *your part.* Negatively charged. **Cathode = the part = where the alloy lands.**
- **The anodes** — usually **inert (insoluble) anodes** (e.g., nickel-coated steel), on the positive side. Unlike acid zinc, they do **not** dissolve to feed the metal.
- **The electrolyte** — the bath: caustic water carrying zinc (as zincate) and nickel (held by amine complexors). The highway both metals travel on.

☀ **FUN FACT**

In acid zinc the anodes dissolve to refill the metal automatically. Zinc-nickel can't do that — there's no anode that dissolves in the right 12–16% proportion — so the metals are fed *separately* and the bath is *actively managed.* It's more work, and it's exactly why zinc-nickel buys you corrosion performance plain zinc can't touch.

WHY ALKALINE ZINC-NICKEL? (AND WHY IT'S LESS FORGIVING)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Outstanding corrosion protection	720–1,000+ hr salt spray with passivate + seal — the whole reason to run it
Uniform alloy & thickness	Consistent %Ni and coverage, even into recesses
Thermal stability	Holds protection after baking and service heat
Hard, wear-resistant deposit	Tougher than plain zinc
No cyanide	Modern alkaline ZnNi is cyanide-free

The flip side — **this bath has little tolerance for sloppiness:**

- **Low cathode efficiency (~50–65%)** → **slow plating** (20–90 min cycles) and lots of hydrogen.
- **Alloy composition must stay in the 12–16% Ni window** — sensitive to temperature, CD, and ratio.
- **Complex chemistry** — two metals + caustic + complexors + brighteners, all actively dosed.
- **Higher hydrogen-embrittlement risk** on hard steel; mandatory bake.
- **Expensive** — nickel and complexors aren't cheap; waste treatment is harder.

★ **REMEMBER**

The headline number for this tank isn't efficiency (it's *low* here) — it's the **alloy: 12–16% nickel.** Everything you do at this station serves one goal: lay down a uniform alloy *in that window.* As the tagline goes: *"Plain zinc protects; the 12–16% nickel makes the protection last."*

⚠ **HEADS-UP (SAFETY & EQUIPMENT)**

Never let bare mild steel parts or tools drop into this bath where they don't belong, and never reach into an energized tank. The bath is **hot caustic with nickel and amine complexors** — full PPE and ventilation. The rectifier is **high amperage** — LOTO before any work.

• THE BATH RECIPE: WHAT'S IN THE TANK AND WHY

The alkaline zinc-nickel bath balances five kinds of ingredients. Know what each *does* and you'll know what goes wrong when it drifts.

1. Zinc (half the alloy). Supplied as **zincate** (zinc dissolved in caustic). Rack/barrel: ~6–12 g/L. Too low → burning and low %Zn balance; too high → pushes the alloy off-ratio and can lower %Ni below the window.

2. Nickel (the other half — and the headline). Held in solution by **amine complexors.** ~0.6–1.6 g/L. This is what makes it zinc-nickel. Too low → %Ni drops below the window (loses corrosion benefit); too high → %Ni climbs above the window (less sacrificial, stress/passivation problems).

3. Sodium hydroxide (caustic — dissolver and conductor). ~100–140 g/L. Dissolves zinc as zincate and carries current. The **Zn:Ni ratio** and the **caustic level** together steer the alloy.

⚙ **TECHNICAL STUFF — THE ZN:NI RATIO IS YOUR ALLOY LEVER**

The bath runs roughly **8:1 to 12:1 zinc-to-nickel by metal.** Move that ratio and you move the *deposit's* %Ni (remember anomalous codeposition — the deposit doesn't mirror the bath). More nickel in the bath → more nickel in the deposit. The chemist adjusts the ratio (and you control temperature) to hold 12–16% Ni. **Know your shop's target ratio.**

4. Amine complexor system (keeps nickel alive). Per TDS. Without complexors, nickel would precipitate as sludge at high pH. They also strongly influence *how much* nickel codeposits. Complexors degrade over time (especially at inert anodes) and are replenished per the supplier's program.

5. Brightener / additive package (appearance + distribution). Per TDS, dosed by amp-hours. Controls grain, brightness, and helps keep the alloy uniform across current densities.

★ **REMEMBER**

A simple memory hook for the four big bath numbers: **Zinc ~6–12, Nickel ~0.6–1.6, NaOH ~100–140, Zn:Ni ~8:1–12:1.** Keep those in your head and you can sanity-check an analysis sheet at a glance — but the *deposit* target is the one that ships: **12–16% Ni.**

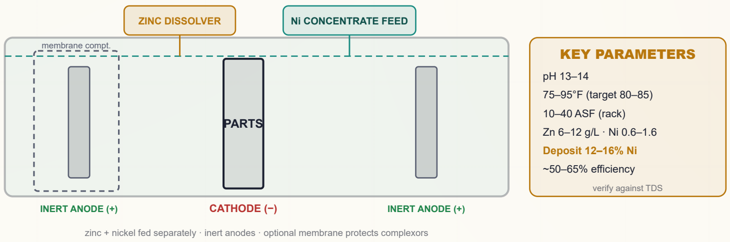
⚠ **HEADS-UP — CARBONATE BUILDUP**

Like all caustic baths, zinc-nickel slowly absorbs CO_2 from air and forms **sodium carbonate**, which accumulates and eventually hurts conductivity and deposit quality. It's monitored and removed (chilling/precipitation) on a schedule — flag it if a reading climbs.

ANODES, THE SEPARATE METAL FEED, AND THE MEMBRANE CELL

This is what makes zinc-nickel hardware different. Know your shop's setup:

- **Inert (insoluble) anodes** — don't dissolve to feed metal; they just carry current. Often nickel-coated steel or specialty coated anodes.
- **Separate zinc feed** — a side **zinc dissolver** (caustic dissolving zinc metal) tops up zincate.
- **Separate nickel feed** — a **complexed nickel concentrate** is dosed to hold the Zn:Ni ratio.
- **Membrane (divided) cell (some shops)** — the anode sits in its own compartment behind an **ion-selective membrane**, protecting the main bath's amine complexors from being oxidized at the anode and improving alloy stability.



⚙ TECHNICAL STUFF — WHY THE MEMBRANE MATTERS

With inert anodes, an unwanted oxidation reaction at the anode surface can slowly break down the amine complexors (and generate byproducts in trace amounts). A membrane keeps that reaction isolated in the anode compartment, so the main bath's complexors — and your alloy control — stay stable longer. If your line has membranes, never run a compartment dry or with a torn membrane.

SUPPORTING EQUIPMENT (AND WHY EACH PIECE EXISTS)

EQUIPMENT	SETTING / SPEC	WHY IT'S THERE
Anodes	Inert; membrane optional	Carry current without throwing off the alloy
Metal feed	Zinc dissolver + complexed Ni concentrate	Replaces the "dissolving anode" function separately
Agitation	Air (oil-free) + cathode-rod; barrel 4–8 RPM	Fresh chemistry at the surface; even alloy; clears H_2
Filtration	Continuous, 2–4 turnovers/hr; carbon as scheduled	Removes particles and degraded organics
Temperature	**75–95°F, target ~80–85°F**	**Drives %Ni** — too hot raises nickel; too cold dulls & lowers %Ni
Current density	1–4 A/dm² (10–40 ASF) rack; 0.3–1.2 barrel	How hard you're driving; CD affects %Ni and burning
Rectifier	High amperage DC; long cycles	Slow plating (low efficiency) — plan time

⚙ **TECHNICAL STUFF**

"Current density" is current spread over the part's surface area (A/dm² or ASF) — *not* total amps. Edges and points see higher CD than flat faces. In zinc-nickel, CD doesn't just affect brightness — **it shifts the local %Ni**, so high-CD edges and low-CD recesses can read slightly different alloy. That's why critical parts are XRF-checked at specified spots.

★ **REMEMBER — TEMPERATURE IS THE ALLOY'S HIDDEN STEERING WHEEL**

Of all the knobs, **temperature moves %Ni the most in daily operation.** A tank drifting a few degrees warm all afternoon can walk the nickel above 16%. Check and hold temperature; report drift.

• YOUR BEST FRIEND: THE HULL CELL (AND THE XRF)

Before dumping chemistry into a production tank on a hunch, test the theory in a tiny tank first — the **Hull Cell**, a small wedge-shaped test tank. You plate one panel under standard conditions (commonly **267 mL at 1–2 A for ~10 min** — follow your TDS, since ZnNi uses lower current than acid zinc). Because the cell is wedge-shaped, the panel sees high CD at one end and low CD at the other at once — an X-ray of your bath across the whole CD range. Read it against your supplier's reference panels.

But zinc-nickel adds a second essential tool: the **XRF analyzer**, which reads the **actual %Ni** (and thickness) on the panel or the part. A Hull Cell tells you about appearance and coverage; **XRF tells you whether you're in the 12–16% window.** On zinc-nickel you use both. (Full XRF section in Part Four.)

WHAT YOU SEE / READ	WHAT IT'S TELLING YOU
Bright, even panel + %Ni 12–16%	Bath in balance — you're good
%Ni reading below 12%	Nickel too low in bath, or temp too low → raise Ni / check temp (chemist)
%Ni reading above 16%	Nickel too high, or temp too high → lower Ni / cool the bath (chemist)
Dull / dark at high CD	Brightener low; CD too high; off-balance metal
Hazy / rough overall	Organic or metallic contamination; carbonate high
Skip/bare patches	Cr(VI) drag-back from passivate, or poor cleaning

💡 **TIP**

Run a Hull Cell *and* an XRF check before and after any chemistry adjustment — before to diagnose, after to confirm the fix landed in the window — on a test panel instead of risking the whole production tank. It's the cheapest insurance on the line.

• METALLIC & OTHER CONTAMINATION

Zinc-nickel is sensitive to contaminants that disturb the alloy or appearance. The biggest unique one:

⚠ **HEADS-UP — HEXAVALENT CHROMIUM DRAG-BACK IS THE CLASSIC ZNNI BATH KILLER**

Cr(VI) that drifts *backward* from the passivate (Station 08) into the zinc-nickel bath causes **bare spots, blistering, and skip-plating** — and it's hard to clear. Prevent it with good rinsing between plating and passivate and by keeping racks/flow moving the right direction. Removal (reduce Cr(VI) → Cr(III), raise pH to precipitate, filter out) is done **only under your chemist's direction.**

Other contaminants — iron, copper, and organics — cause haze, dark deposits, and dullness; managed by tight rinsing, filtration, carbon treatment, and dummying, per your chemist. Carbonate is managed by chilling/precipitation on schedule.

STANDARD OPERATING PROCEDURE (STATION 05)

1. **Confirm the part is clean and active** — straight from the pre-plate rinse, into plating **within ~30 seconds.** A part that sat and oxidized plates unevenly. If in doubt, send it upstream.
2. **Check bath readiness:** temperature **75–95°F (target 80–85°F)**; caustic level in range; **Zn:Ni ratio** on target per the latest analysis; agitation running (air + cathode rod, or barrel 4–8 RPM); filtration running; anodes/membrane compartment in good order; zinc dissolver and nickel feed functioning.
3. **Verify chemistry is current** — zinc, nickel, caustic, complexors, and brighteners analyzed and replenished on schedule (per TDS / amp-hours), and the **last %Ni reading was in window.**
4. **Set the rectifier** to the correct current for the part's surface area and target CD. Expect a **long cycle** (20–90 min) because efficiency is low.
5. **Load the part / start the cycle.**
6. **Monitor while running** — temperature especially (it steers %Ni), plus agitation, filtration, color, and gassing.
7. **At cycle end, remove and drain over the tank,** then transfer promptly to the post-plate rinse (Station 06).
8. **Log it** — amp-hours, additions, temperature, last %Ni. This log is how you keep the alloy in window tomorrow.

⚠ **HEADS-UP (HYDROGEN EMBRITTLEMENT — READ THIS TWICE)**

Zinc-nickel is a **low-efficiency, high-hydrogen** process. On **hardened steel (above 40 HRC / ~180 ksi)**, absorbed hydrogen can crack parts later without warning. Such parts **must be baked** after plating (Station 06): **375 ±25°F, ≥ 8 hours, within 4 hours of plating (ASTM B850).** Never skip it on embrittlement-sensitive work.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE / READ	MOST LIKELY CAUSE	WHAT TO DO
%Ni below 12%	Low nickel; temp too low; CD too high	Chemist: raise Ni / Zn:Ni; check/raise temp
%Ni above 16%	High nickel; temp too high; CD too low	Chemist: lower Ni; cool bath
Dull / dark deposit	Brightener depleted; off-balance metal; organics	Add brightener per amp-hour; carbon treat; rebalance
Burning at high CD (rough edges/tips)	CD too high; zinc/caustic off; agitation poor	Lower CD; check ratio; improve agitation
Bare/skip spots, blisters	**Cr(VI) drag-back** from passivate; poor cleaning	Stop drag-back; verify cleaning; chemist removes Cr
Hazy / rough	Iron/organics; carbonate high; poor filtration	Analyze; carbon treat; remove carbonate; check filter
Peeling / poor adhesion	Inadequate clean/activation; oxidized before plating	Look upstream (Stations 1–4); speed the handoff
Uneven %Ni across the part	Wide CD variation; agitation/temperature variation	Check fixturing/agitation; XRF at spec locations

• SAFETY — THIS STATION DEMANDS YOUR RESPECT

⚠ **HEADS-UP — CAUSTIC + NICKEL + AMINE MIST**

The bath is hot caustic with nickel and complexors; gassing throws a fine mist you must not breathe — local exhaust ventilation mandatory. **Nickel mist is a respiratory sensitizer.** Full PPE: splash goggles *and* face shield, elbow-length chemical gloves, apron, boots, respirator per shop. Bath on skin → flush 15 minutes and report.

⚠ **HEADS-UP — THE RECTIFIER IS HIGH AMPERAGE**

Shock and arc-flash risk. **LOTO** before any maintenance, reaching into the tank, or bus-bar work. No exceptions, no matter how quick.

⚠ **HEADS-UP — MEMBRANE / ANODE COMPARTMENT**

If your line has a membrane cell, never run the anode compartment dry, overfilled, or with a damaged membrane — it can crash your alloy control and damage equipment. Report anything off about the anolyte level or color.

• TEACH-BACK & TOP MISTAKES — STATION 05

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge?
- The four bath numbers (zinc, nickel, caustic, Zn:Ni ratio) and the *deposit* target (12–16% Ni)?
- What anomalous codeposition means (deposit ≠ bath ratio)?
- Why the anodes are inert and the metals fed separately?
- Why temperature is the biggest everyday driver of %Ni?
- What a membrane compartment does and why?
- Why Cr(VI) drag-back is so damaging here?
- What XRF measures and why both Hull Cell and XRF are used?
- Why HE/baking is mandatory on hard steel in this low-efficiency bath?

TOP MISTAKES TO AVOID

- Trusting the bath ratio instead of measuring %Ni.
- Letting temperature drift (walks the alloy out of window).
- Ignoring rising carbonate.
- Allowing Cr(VI) drag-back from passivate.
- Plating a part that sat too long and oxidized.
- Expecting acid-zinc speed (this bath is slow — plan time).
- Skipping the embrittlement bake on hard steel.

SIGN-OFF — STATION 05

I confirm I can independently verify bath readiness (temp, caustic, Zn:Ni ratio, agitation, filtration, anodes/membrane, metal feeds); read an analysis sheet and an XRF %Ni result against the 12–16% window; run and interpret a Hull Cell panel; recognize the main contaminants (esp. Cr(VI) drag-back) and their sources; and state the hydrogen-embrittlement bake requirement for hardened steel. I understand chemistry adjustments are made only by authorized personnel.

Operator ________________ Date ____________ Trainer ________________ Date ____________

STATION 06

POST-PLATE RINSE & THE BAKE DECISION

WASH THE CAUSTIC AND NICKEL OFF — AND DECIDE WHETHER A PART GOES TO THE OVEN

Your part just came out of a caustic, nickel-loaded plating bath, dripping with both. The next tank is the nitric/acid activation, then the passivate. Drag caustic and nickel forward and you waste expensive chemistry, neutralize the acid activation, and contaminate the passivate. This rinse is the wall between the high-pH plating bath and the low-pH finishing tanks. It's also the decision point for whether a hardened part goes to an *oven* first.

> ★ **REMEMBER**
> A rinse is never "just a rinse." Each one is a *checkpoint* that protects the next tank. This one protects the finishing tanks from caustic and nickel — and recovers expensive nickel if you run it as a recovery rinse.

• WHAT AND WHY

A thin film of plating bath clings to the part — full of **caustic and nickel.** If it rides into the finishing tanks, three bad things happen: (1) the nitric/passivate chemistry is **neutralized/contaminated**; (2) conversion coatings come out **thin or blotchy**; (3) **bath life drops** — and you lose costly nickel down the drain.

> 💡 **TIP — DRAG-OUT RECOVERY (THE MONEY-SAVER)**
> Many shops make the *first* post-plate rinse a **still "dead" recovery rinse.** Its nickel-bearing, mildly caustic water is periodically pumped back to top off the plating bath — recovering expensive nickel and cutting waste-treatment load. After the recovery rinse, a flowing rinse polishes the part clean.

```
RECOVERY + COUNTER-FLOW RINSE  ( parts -> )

from PLATING   RECOVERY (still)    RINSE 1        RINSE 2     to NITRIC
(Ni + caustic) [dead tank]  ----> [dirtier] ---> [cleaner] ->  ACTIVATION
                   |                  ^               |
     pump nickel   |                  | overflow back |
     back to ------+                  +--- WATER <----+ fresh DI in
     PLATING (saves nickel, cuts waste)
```

• OPERATOR ESSENTIALS

PARAMETER	TARGET	WHY IT'S SET THERE
Conductivity (final flowing rinse)	< 200 µS/cm (aim < 100)	Caustic/nickel drag-in is the whole concern
Temperature	Ambient	No heating needed
Time	30–60 sec with agitation	Flush the film; **double rinse** recommended
Drain time	15–20 sec over the tank	Excess water dilutes finishing tanks
Water	DI preferred	Avoids hard-water haze on fresh alloy
Recovery rinse	Still tank, pumped back to plating	Recovers nickel; cuts waste

• STANDARD OPERATING PROCEDURE

1. **Transfer promptly from plating.** Don't let the plated part sit and oxidize.
2. **Recovery rinse first (if used),** then **flowing rinse with agitation, 30–60 sec.** Flush blind holes and recesses.
3. **Move dirty → clean** if double-rinsing, so final contact is the cleanest water.
4. **Drain 15–20 seconds** over the tank.
5. **Handle with clean gloves only** — bare fingers leave oils/salts on the fresh alloy.
6. **Move promptly** to the nitric activation (Station 07) — or to the **bake** if the spec requires baking before finishing.

THE HYDROGEN-EMBRITTLEMENT BAKE (CRITICAL FOR HARD STEEL)

> ⚠ **HEADS-UP (SAFETY-CRITICAL — PREVENTS CATASTROPHIC PART FAILURE)**
> Zinc-nickel is a **low-efficiency, high-hydrogen** process, so embrittlement risk is real on hard steel. For steel **above 40 HRC**: bake **within 4 hours** of the last electroplating step, at **375 ±25°F**, for a **minimum of 8 hours**, per **ASTM B850.** Some customer specs require the bake *before* passivation — check the spec. A missed bake on a structural fastener isn't a cosmetic defect; it's a part that can snap in service. Never skip it on embrittlement-sensitive work.

> ⚙ **TECHNICAL STUFF**
> Zinc-nickel handles the bake *better* than plain zinc — its corrosion performance survives 375°F far better than plain zinc's would. That's a genuine advantage: you can do the mandatory embrittlement relief bake without sacrificing the corrosion protection you just built. (The chromate/seal heat limits still apply at Stations 08–09; this bake happens before those films go on.)

TROUBLESHOOTING (STATION 06)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Finishing-bath life short	Caustic/nickel drag-in from this rinse	Tighten conductivity; add a stage; use recovery rinse
Blotchy passivate later	Residual caustic/nickel on the alloy	Improve rinse agitation and time
White haze on the alloy	Oxidizing, or hard-water deposits	Use DI; reduce transfer time
Nickel in effluent high	No recovery rinse; over-flowing first stage	Add recovery rinse; pump back to plating

TOP MISTAKES AT STATION 06

1. **Treating it as optional** — it sets finishing-bath life and recovers nickel.
2. **Letting conductivity climb** above 200 μS/cm without more flow.
3. **No drain time** — flooding/diluting the finishing tanks.
4. **Skipping the recovery rinse** — pouring costly nickel down the drain.
5. **Skipping or delaying the embrittlement bake** on hardened steel.

TEACH-BACK CHECKLIST

- Explain why caustic/nickel drag-in matters at this rinse.
- State the conductivity target, and what μS/cm measures.
- Explain a drag-out recovery rinse and why it saves money and reduces nickel waste.
- State the embrittlement bake parameters exactly (375 ±25°F, ≥ 8 hr, within 4 hr, > 40 HRC, ASTM B850).
- Explain why ZnNi tolerates the bake better than plain zinc.

SIGN-OFF — STATION 06

I confirm I can measure/interpret rinse conductivity; hold the < 200 μS/cm target; run a recovery + double counter-current rinse; state and apply the ASTM B850 bake (375 ±25°F, ≥ 8 hr, within 4 hr of plating); and handle fresh alloy without contaminating it.

Operator ______________ Date __________ Trainer ______________ Date __________

STATION 07

NITRIC/ACID ACTIVATION

THE BRIDGE BETWEEN PLATING AND PASSIVATING — BRIGHTEN AND WAKE UP THE ALLOY

Fresh zinc-nickel comes out of a high-pH bath with a dull caustic film and a little surface oxidation. The passivate (Station 08) needs a **clean, bright, chemically active alloy surface** to react with evenly. A quick dip in **dilute nitric acid** (or another acid per your TDS) strips that film, brightens the alloy, and "wakes it up" so the chromate builds a uniform, correctly-colored film. Skip or botch this little step and you get blotchy, weak passivates no matter how good the plating was.

> ★ **REMEMBER**
> This dip is short but pivotal. **A poor nitric activation = a blotchy, thin, wrong-colored passivate.** It's the handshake between the alloy and the finish.

• WHAT YOU'RE DOING & WHY IT MATTERS

The acid does two jobs: it **neutralizes and removes the alkaline film** left from plating, and it lightly **micro-etches/brightens** the alloy so the chromate reaction starts uniformly across the surface. On zinc-nickel specifically, a clean activation also helps the passivate develop its full corrosion performance and color (clear/blue or black).

> ⚙ **TECHNICAL STUFF — WHY NITRIC**
> Dilute nitric acid is a mild oxidizer that brightens the zinc-nickel surface and dissolves the light oxide/caustic film without aggressively attacking the alloy at short times. It leaves a clean, slightly activated surface ideal for trivalent chromate. Concentration and time are low and *must be controlled* — too long and you start removing alloy you just plated.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Acid	`Dilute nitric (typ. 0.5–3% v/v) or per TDS`	Some lines use a mild acid blend — follow TDS
Temperature	`Ambient`	No heating
Time	`5–30 sec`	Short — just enough to brighten/activate
Agitation	`Gentle manual movement`	No air (acid mist)
Rinse after	`Brief rinse before passivate (per shop)`	Some lines go straight to passivate — follow process

> ⚠ **HEADS-UP — NITRIC ACID**
> Nitric is a **strong oxidizer and severe burn/fume hazard.** It releases irritating **brown nitrogen-oxide fumes**, reacts dangerously with organics and some metals, and must **never** be mixed with the caustic bath, cyanide (if any is on site), or organic chemistries. **Ventilation required.** PPE: goggles + face shield, **butyl/neoprene** acid gloves, apron, boots, respirator if ventilation is inadequate. Acid-to-water always. Skin/eyes → flush 15 minutes.

STANDARD OPERATING PROCEDURE

1. **Confirm the part came clean** from the post-plate rinse (low caustic/nickel carry-over) and that any required **bake** was done if the spec calls for baking before finishing.
2. **Verify acid concentration and temperature** are in range; ventilation on; PPE on.
3. **Immerse for the short specified time (5–30 sec)** with gentle manual agitation — don't over-dip and remove alloy.
4. **Brief rinse** (if your process includes one) and move **promptly** to the passivate before the bright surface re-oxidizes.

TEACH-BACK CHECKLIST

- Explain the two jobs of this station (remove the alkaline film; brighten/activate the alloy for chromate).
- State the dip time range and why it's short (5–30 sec; over-dipping removes alloy).
- Name the special hazards of nitric (oxidizer, brown fumes, never mix with caustic/organics).
- Explain what a poor activation does to the passivate (blotchy, thin, wrong color).

TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Blotchy / uneven passivate	Weak or skipped activation; caustic drag-in	Verify acid strength/time; improve post-plate rinse
Over-bright/etched, thin alloy	Over-dipped in nitric	Shorten time; reduce concentration
Dull alloy into passivate	Acid spent/too weak	Refresh/strengthen per TDS
Strong brown fumes	Over-concentrated; poor ventilation	Dilute to spec; check exhaust

SIGN-OFF — STATION 07

Trainee ________________ Trainer ________________ Dip used __________ sec

"I confirm I ran the nitric/acid activation at the short specified time with manual agitation only, ventilation and full acid PPE in place, did not over-dip, and moved promptly to passivate."

TIP

If your passivate color keeps coming out uneven, don't start tinkering with the passivate first — check this little dip and the rinse before it. Nine times out of ten a "passivate problem" is really a tired nitric tank or caustic riding through from Station 06.

STATION 08

PASSIVATE — TRIVALENT CHROMATE

THE CHROMATE CONVERSION COATING THAT PROTECTS THE ALLOY

You've made a bright, correctly-alloyed zinc-nickel deposit — but bare alloy still benefits from a protective skin and color, and it's the passivate (plus the seal at Station 09) that takes corrosion life into the premium range. The passivate is a **chromate conversion coating**: dip the part and the solution chemically *reacts with the alloy surface* to grow a thin, tough, protective film. As the tagline goes: *"The alloy is the engine; the passivate and seal are the turbocharger."*

★ **REMEMBER**

The zinc-nickel protects the steel. The passivate protects the alloy. The seal protects the passivate. It's protection stacked on protection — and these last stations are how you reach 1,000-hour salt spray.

WHAT "CONVERSION COATING" MEANS (DIFFERENT FROM PLATING)

Station 05 was **electroplating** — electricity *added* metal. Station 08 is a **conversion coating** — *no electricity.* You dip the part and the solution reacts with the alloy surface, converting the very top layer into a thin chromium-rich protective film. It isn't a separate layer laid on top; it's the alloy surface *transformed.* That's why it's called "conversion."

⚙ **TECHNICAL STUFF**

In the dip, the mild acid in the passivate lightly attacks the surface, and chromium compounds deposit into the reacting surface, building a gel-like chromium-rich film that then dries hard — a stable barrier that slows corrosion. Modern **trivalent Cr(III)** films generally do not "self-heal" the way old hexavalent films did; they protect as a stable barrier, and on zinc-nickel they pair with a seal for top performance.

THE COLORS ON ZINC-NICKEL

Zinc-nickel is most often finished **clear/blue** or **black.** (Yellow/iridescent trivalent is less common on ZnNi than on plain zinc.) The color signals the film and, with the seal, the protection class — but **the customer's spec decides the color, not operator preference.**

TYPE	LOOK	TYPICAL USE ON ZNNI
Trivalent Clear/Blue	Nearly clear, faint blue	Most common; pairs with seal for high salt-spray
Trivalent Black	Black	Common on automotive/fastener ZnNi; needs ZnNi-specific black

💡 **TIP**

Zinc-nickel **black** passivates are a specialty — many were historically **hexavalent.** If your shop runs black ZnNi, *confirm it's trivalent* (or that management has signed off on a controlled hexavalent process). Use a passivate **formulated specifically for zinc-nickel** — plain-zinc passivates do not perform the same on the alloy.

HEXAVALENT VS. TRIVALENT: THE MOST IMPORTANT DISTINCTION

- **Trivalent — Cr(III):** the modern, **RoHS-compliant**, much safer chemistry. The default for zinc-nickel.
- **Hexavalent — Cr(VI):** the legacy chemistry. Builds durable films but is a **known human carcinogen**, *not* RoHS-compliant, and being phased out.

> ⚠ **HEADS-UP (SAFETY — THE SINGLE MOST IMPORTANT WARNING ON THIS STATION)**
> **Hexavalent chromium, Cr(VI), is a known carcinogen.** OSHA PEL is just **5 µg/m³.** Where used at all, it demands a **supplied-air respirator, full protective suit, and medical surveillance per OSHA 29 CFR 1910.1026.** Its waste is **RCRA hazardous. Trivalent is the default — do not substitute hexavalent without full regulatory/safety review and management sign-off.** And remember: Cr(VI) dragged *backward* into the plating bath ruins the alloy deposit (skip/blister).

> ☀ **FUN FACT**
> Getting durable *black* on zinc-nickel without hexavalent chromium was a real chemistry challenge — modern trivalent ZnNi blacks plus seals are a quiet success of green metal finishing, hitting corrosion numbers the old hex blacks reached, without the carcinogen.

OPERATOR ESSENTIALS

The single most important control variable here is **pH** — every passivate has a narrow "critical pH window," and outside it the film comes out too thin, too thick, non-adherent, or the wrong color.

PARAMETER	TRIVALENT CLEAR/BLUE	TRIVALENT BLACK
Temperature	75–95°F	75–110°F
pH	1.6–2.5 (per TDS)	2.0–4.0 (per TDS)
Time	30–60 sec	45–120 sec
Cr type	Cr(III) RoHS	Cr(III) RoHS
Salt spray *with seal* (red rust)	720–1,000+ hrs	600–1,000+ hrs

> ★ **REMEMBER**
> The rule-card number is **"pH per TDS."** The window depends on the specific product — a typical trivalent clear/blue sits around **1.6–2.5**, but *your* TDS is the authority. pH is *the* control knob; check it often.

> ⚙ **TECHNICAL STUFF — WHAT IS "SALT SPRAY"?**
> An accelerated corrosion test (ASTM B117): parts sit in a sealed cabinet sprayed with salt fog; you record the *hours* until corrosion appears. For zinc-nickel, customers often specify both **hours to white corrosion** and **hours to red rust** (steel showing through). The big 720–1,000+ hr numbers are *red-rust* figures achieved with passivate **plus seal** — which is why Station 09 matters so much.

STANDARD OPERATING PROCEDURE (STATION 08)

1. **Confirm the part came clean and bright** from the nitric activation (Station 07) with low carry-over.
2. **Confirm any required embrittlement bake was done** if the spec calls for baking before passivation.
3. **Check the passivate bath:** correct type for the spec, **pH in the product's window** (verify against TDS), temperature in range, and that it's a **ZnNi-specific** product.
4. **Immerse for the specified time** (e.g., 30–60 sec clear/blue; 45–120 sec black). Time and pH together control film thickness and color.
5. **Drain over the tank,** then proceed to the **seal/topcoat and final rinse/dry — Station 09.**

• TROUBLESHOOTING (STATION 08)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Thin / pale film	pH too high; bath depleted	Lower pH; replenish per TDS
Wrong color / weak black	pH off; wrong (plain-zinc) passivate; weak activation	Check pH; use ZnNi-specific product; fix Station 07
Blotchy / uneven	Caustic/nickel drag-in; uneven alloy underneath	Improve post-plate rinse; check plating uniformity/%Ni
Film rubs off	pH out of range; insufficient cure	Check pH; allow full cure before handling
Bath life short	Caustic/metal drag-in	Improve Stations 06–07 rinses

• INTEGRATED SAFETY (STATION 08)

⚠ **HEADS-UP (TRIVALENT)**

Even “safe” trivalent passivate is **acidic (pH 1.6–4) and causes burns.** Goggles + face shield, chemical gloves, apron. Flush skin/eyes 15 minutes on contact.

⚠ **HEADS-UP (HEXAVALENT — WHERE STILL USED, ESP. SOME BLACKS)**

Known carcinogen. OSHA PEL 5 µg/m³. Supplied-air respirator, full suit, medical surveillance per **OSHA 29 CFR 1910.1026.** Waste is **RCRA hazardous.** Strong recommendation: run trivalent.

• TEACH-BACK & TOP MISTAKES — STATION 08

TEACH-BACK — CAN YOU ANSWER?

- Why does plated zinc-nickel get a passivate — what is “white rust”?
- How is conversion coating different from electroplating?
- Cr(III) vs. Cr(VI) — key safety/regulatory facts for each?
- Why pH is the critical control variable; the clear/blue window?
- Why use a passivate formulated *for zinc-nickel* (and confirm blacks are trivalent)?
- How does Cr(VI) drag-back hurt the plating bath?

TOP MISTAKES TO AVOID

- Letting pH drift out of the window.
- Carrying caustic/nickel in from a weak Station 06 rinse.
- Using a plain-zinc passivate on the alloy.
- Assuming a black passivate is trivalent without checking.
- Allowing Cr(VI) to drag back into the plating bath.
- Handling/testing before the cure.

SIGN-OFF — STATION 08

I confirm I can distinguish trivalent from hexavalent chromium and state the hazard/PPE for each; select and verify the correct ZnNi-specific passivate type/color for the spec; hold pH within the product’s critical window; diagnose common film defects and trace them upstream; and prevent Cr(VI) drag-back.

Operator ____________ Date ____________ Trainer ____________ Date ____________

STATION 09

SEAL / TOPCOAT, FINAL RINSE & DRY

THE LAST STAGE — WHERE ZINC-NICKEL REACHES 720–1,000+ HOURS

The passivate is on. One stage remains, and on zinc-nickel it's the one that earns the premium corrosion numbers: the **seal/topcoat.** Station 09 has three jobs: **apply a seal** that boosts salt-spray life (and often adds controlled lubricity for fasteners), **rinse off leftover chemistry**, and **dry/cure gently** so the soft fresh films set without cracking. It feels like the easy part — which is exactly why parts get wrecked here.

> ★ **REMEMBER**
> The passivate and seal are **soft when fresh.** Heat cracks them; touching rubs them off. Patience at Station 09 protects every step upstream — and the 1,000-hour result you're being paid for.

WHAT YOU'RE DOING & WHY IT MATTERS

A **seal (topcoat)** is a thin organic or inorganic layer applied over the passivate that fills micro-pores, adds barrier protection, and (on fasteners) delivers a controlled **coefficient of friction** so bolts torque consistently. Sealing is what typically takes zinc-nickel from a few-hundred-hour passivate to **720–1,000+ hours** to red rust. Many seals are applied warm and then **cured** at a controlled temperature.

> ⚙ **TECHNICAL STUFF — SEAL VS. PASSIVATE**
> The passivate is a chromium-rich conversion film grown *into* the surface; the seal is an additional layer applied *on top*. They work together: passivate provides the chemically bonded chromium barrier, seal provides extra physical barrier, pore-filling, and friction control. Cure temperature and time are set by the seal's TDS — too cool and it doesn't cure, too hot and you risk cracking the passivate underneath.

OPERATOR ESSENTIALS

STEP	SETTING	WHY IT'S SET THERE
Final rinse (pre-seal, if used)	`DI, ambient, 15–30 sec`	Removes passivate residue without stripping the film
Seal/topcoat	`Per TDS (often warm dip, then cure)`	Boosts salt spray; adds friction control
Cure temperature	**`Per seal TDS — commonly ≤ 175–200°F`**	Too hot cracks the passivate beneath; follow TDS
Cure / dry time	`Per TDS; then` **`24 hr cure`** `before testing/packing`	Soft films need time to harden
Handling	`Clean gloves only`	Bare hands mar soft films, leave salts/oils

> ⚠ **HEADS-UP (DON'T RUIN A GOOD FINISH AT THE LINE)**
> Two easy ways to destroy a perfect zinc-nickel finish in the last minute: **drying/curing too hot** (cracks the passivate and degrades corrosion life — follow the seal TDS limit) and **handling, testing, or packing before the full cure** (the soft film rubs off and salt-spray results read falsely low). Patience here protects everything you did upstream.

STANDARD OPERATING PROCEDURE (STATION 09)

1. **Final DI rinse** (if your process includes one) — gentle immersion, ambient, 15–30 sec.
2. **Apply the seal/topcoat** per TDS (often a warm dip), then drain.
3. **Cure/dry** at the **seal's specified temperature** — never hotter, or you crack the films and lose corrosion life.
4. **Set aside to cure** — at least **24 hours** before packing, salt-spray testing, or rough handling.
5. **Handle with clean gloves** throughout; inspect for spotting, haze, correct color, and (where required) verify thickness/%Ni by XRF per the spec.

TEACH-BACK CHECKLIST

- State the three jobs of Station 09 (seal for corrosion/friction; rinse residue; gently cure/dry).
- Explain why the seal is what gets zinc-nickel to 720–1,000+ hours.
- State the cure temperature rule and what happens if you exceed it (per seal TDS; too hot cracks the passivate / loses corrosion life).
- State the cure time before packing/testing and why (24 hr min; soft films must harden).
- Explain why fresh, uncured parts are handled with clean gloves only.

TOP MISTAKES NEW OPERATORS MAKE

1. **Curing too hot to "speed things up."** You crack the films you just built and lose the corrosion life.
2. **Packing or testing before the cure.** Soft films rub off; salt-spray reads falsely low.
3. **Skipping the seal** when the spec calls for it — you'll miss the salt-spray requirement.
4. **Bare-handed handling** of fresh, soft films — fingerprints and mars.

SIGN-OFF — STATION 09

Trainee ____________________ Trainer ____________________ Date ____________________

"I confirm the part received its final rinse and seal/topcoat, was cured/dried at or below the seal's specified temperature, was handled with clean gloves, and was set aside for the full cure before packing or testing — with required PPE worn."

★ REMEMBER

You've now walked the whole nine-stage line, tank by tank. The part that ships at 1,000 hours is the sum of every station doing its small job right — a clean surface, tight rinses, an in-window alloy, a clean activation, an even passivate, and a patient cure. That's the craft.

PART FOUR — ACROSS-THE-LINE & BACK MATTER
THE DEFINING SKILL

ALLOY COMPOSITION CONTROL & MEASURING %NI

THE SKILL THAT DEFINES A ZINC-NICKEL OPERATOR

On a plain-zinc line you control thickness and appearance. On zinc-nickel you control those **plus the alloy composition** — and composition is what the customer is really buying. This section pulls together everything about keeping the deposit in the **12–16% Ni** window and how you *prove* it.

• WHY COMPOSITION IS QUALITY-CRITICAL

Corrosion performance, thermal stability, and even how well the passivate takes all depend on the alloy being in the gamma-phase window (~12–16% Ni). Out of window:

- **Below ~10–12%:** behaves more like plain zinc — fails the salt-spray spec.
- **Above ~16–18%:** less sacrificial (may stop protecting the steel at scratches), more internal stress, passivation problems.

So composition isn't a "nice to have" — it's a **pass/fail spec parameter**, checked and recorded like thickness.

• MEASURING %NI: XRF IS YOUR INSTRUMENT

X-Ray Fluorescence (XRF) is the standard tool. It's a non-destructive bench (or handheld) instrument: it shines X-rays at the coating, the coating's atoms fluoresce back at characteristic energies, and the analyzer reads both the **coating thickness** and the **%Ni in the alloy** in seconds.

How you use it:

- **Calibrate/verify** against known standards per your shop's schedule and the spec.
- **Measure at the spec-defined locations** — critical parts call out *where* to read (flat faces vs. high/low current-density areas), because %Ni varies a little with current density.
- **Record** thickness and %Ni on the job/quality record.
- **Trend it** — a slow climb or drop in %Ni over days is your early warning that temperature or the Zn:Ni ratio is drifting.

> ⚙ **TECHNICAL STUFF — XRF BASICS**
> Every element fluoresces X-rays at its own "fingerprint" energy. The analyzer counts the zinc and nickel fluorescence and computes the ratio (%Ni) and the total coating thickness. It's fast and non-destructive, which is why it's run right on the floor. It does need correct calibration standards and proper geometry — garbage setup, garbage reading.

> ★ **REMEMBER**
> **The bath ratio predicts the alloy; XRF proves it.** Never certify a zinc-nickel job on bath chemistry alone. The %Ni reading is the number the customer's spec lives or dies on.

> 💡 **TIP**
> If %Ni trends out of window with no chemistry change, suspect **temperature** first, then current density / fixturing, then the Zn:Ni ratio. Report the trend to your lead/chemist — operators flag it; the chemist corrects the bath.

• HYDROGEN EMBRITTLEMENT & BAKING (LINE-WIDE)

Because zinc-nickel is low-efficiency and uses acids in prep, hydrogen loads into steel at Stations 03 and 05. On **high-strength steel (> 40 HRC / > 180 ksi):**

- **Bake at 375 ±25°F, ≥ 8 hours, within 4 hours of plating** (ASTM B850; specific specs may differ — follow the print).
- Prefer **anodic** electrocleaning and **inhibited, minimum-time** pickling to reduce uptake in the first place.
- Zinc-nickel **tolerates the bake well** — its corrosion performance survives 375°F far better than plain zinc, so you get the safety benefit without losing protection.

> ⚠ **HEADS-UP**
> The bake is **not optional** on embrittlement-sensitive parts. A missed bake can mean a structural part that cracks in service. When a traveler flags hardness, baking is a safety control, not a quality nicety.

• RINSING & WATER

Zinc-nickel rinses do double duty: contamination control *and* **nickel recovery** (drag-out recovery rinses returned to the plating bath save costly nickel and cut effluent load). Use **DI water** near the plating and finishing tanks, run **counter-current** rinses, and hold the tight conductivity targets (< 200 µS/cm at the guard and post-plate rinses).

• WASTE & ENVIRONMENTAL (NICKEL + AMINES + CHROMIUM)

Zinc-nickel effluent is more complex than plain zinc:

- **Nickel** is a regulated heavy metal — recover it (recovery rinses, ion exchange) and treat to permit limits.
- **Amine complexors** can hold metals in solution and resist breakdown, complicating conventional hydroxide precipitation; specialized treatment may be required (follow your facility's wastewater program).
- **Chromium** from passivate must be treated (Cr(VI) reduced to Cr(III), then precipitated) — and never let Cr(VI) reach the plating bath.
- Manage all per **EPA**, local POTW permits, and your facility's plan. Operators: never send concentrated drag-out or spills to a floor drain.

> ⚠ **HEADS-UP**
> Because amine complexors can re-dissolve metals you're trying to remove, **don't improvise waste handling.** Keep streams segregated as your shop specifies, and report anything unusual to your environmental lead.

• QUALITY CHECKS (THE OPERATOR'S TOOLKIT)

CHECK	TOOL	WHAT IT CONFIRMS
Coating thickness	XRF	Meets spec thickness (8–12 µm typical)
Alloy %Ni	**XRF**	**In the 12–16% window**
Appearance/coverage	Visual + Hull Cell	Bright, even, no skip/burn
Adhesion	Bend/tape per spec	No peeling (upstream prep verified)
Corrosion	Salt spray ASTM B117	Hours to white/red rust meet spec
Bath balance	Titration / analysis	Zn, Ni, NaOH, ratio, carbonate in range

> ★ **REMEMBER**
> Most zinc-nickel defects trace to **four root causes:** alloy out of window (temp/ratio), contamination (Cr(VI) drag-back, iron, organics, carbonate), a weak rinse letting drag-in through, or poor prep (clean/activation). Diagnose with **XRF + Hull Cell** before dumping chemistry, and always look *upstream*.

PLAIN-LANGUAGE REFERENCE

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

> **TIP**
> Keep this glossary tabbed. The fastest way to sound like a veteran is to use the right word for the right thing.

Adhesion: How well the plated alloy sticks to the steel. Good adhesion means it won't peel.

Agitation: Movement of solution or parts; keeps fresh chemistry at the surface, clears hydrogen bubbles (which cause pits), evens the deposit and the alloy. Air, mechanical, or barrel rotation.

Alkaline: The opposite of acidic; high pH (above 7). The cleaner and the zinc-nickel bath are strongly alkaline (caustic).

Alloy: A mixture of metals at the atomic level — here zinc + nickel.

Amine Complexor: Organic molecules that "wrap around" nickel and keep it dissolved in the high-pH bath; they also strongly affect how much nickel codeposits. An exposure and waste-treatment concern.

Ambient: Room temperature, ~60–85°F. No heating needed.

Amp-Hour (A·h): "How much plating you've done": amps × hours. Brighteners and additives are dosed per amp-hour.

Anode: The positively-charged side. In alkaline ZnNi these are usually **inert (insoluble) anodes** that carry current but don't feed metal.

Anomalous Codeposition: The quirk where the *less* noble metal (zinc) plates preferentially, so the deposit's %Ni (12–16%) is far lower than the bath's zinc:nickel ratio would suggest.

ASF: Amps per square foot — a current-density unit. ZnNi rack runs ~10–40 ASF.

ASTM B117: The neutral salt-spray (salt fog) corrosion test standard.

ASTM B841: A common specification for zinc-nickel alloy coatings.

ASTM B850: The standard for the hydrogen-embrittlement relief bake.

Barrel Plating: Plating many small parts together in a rotating perforated barrel. ZnNi barrel cycles run long (slow process).

Brightener System: Organic additives that make the deposit bright and help keep the alloy uniform; dosed per TDS.

Carbonate (Na_2CO_3): Builds up in caustic baths from absorbed CO_2; eventually must be removed (chilling/precipitation).

Carbon Treatment: Circulating the bath through activated carbon to remove organic contamination and degraded additives.

Cathode: The part being plated — the negatively-charged work. The alloy deposits on the cathode.

Cathode Efficiency: The percentage of current that actually deposits metal (vs. making hydrogen). Alkaline ZnNi is **low — ~50–65%** — which is why it plates slowly and why hard parts must be baked.

Caustic / Caustic Soda: Sodium hydroxide (NaOH). Strongly alkaline; severe burn hazard.

Codeposition: Two (or more) metals plating onto the same part at the same time, from the same bath — here zinc + nickel.

Complexor / Complexing: Chemically "holding onto" metal ions so they stay dissolved and available — here, amines holding nickel at high pH.

Conductivity (µS/cm): A measure of dissolved salts in rinse water — how "dirty" the rinse is. **Lower is cleaner.**

Counter-Current Rinse: A water-saving setup where fresh water flows opposite to the parts, so the cleanest water meets the cleanest parts.

Cr(III) / Trivalent Chromium: The modern, RoHS-compliant chromium in today's passivates. Much safer than hexavalent.

Cr(VI) / Hexavalent Chromium: The old form. A **known carcinogen**, NOT RoHS-compliant. Legacy use only, under strict OSHA controls. Also the classic ZnNi bath contaminant when it drags *back* into plating.

Current Density (CD): Current per unit of part surface area (ASF or A/dm²). High-CD areas (edges) and low-CD areas (recesses) behave differently — and in ZnNi they can have slightly different %Ni.

• GLOSSARY (CONTINUED)

DI Water (Deionized Water): Water with dissolved minerals removed. Cleaner than tap; preferred near the plating and finishing tanks.

Drag-In / Drag-Out: Chemistry riding on parts from one tank to the next. *Drag-out* leaves a tank; *drag-in* arrives in the next. Rinses control it. (Recovery rinses turn ZnNi drag-out into recovered nickel.)

Drag-Out Recovery (Dead) Rinse: A still rinse whose nickel-bearing water is pumped back to the plating bath to recover metal and cut waste.

Dummy / Dummying: Plating onto scrap "dummy" cathodes at low current to draw contaminating metals out of the bath.

Electroclean: Cleaning with electrical current applied (anodic preferred for hard steel) for a truly spotless surface.

Filtration: Continuously pumping the bath through a filter (2–4 turnovers/hr) to remove particles.

Gamma (γ) Phase: The protective zinc-nickel crystal structure formed in the ~12–16% Ni window — the source of the corrosion performance.

Hexavalent: See Cr(VI).

Hull Cell: A small, wedge-shaped test tank that plates a panel across a *range* of current densities at once; reading the panel (and XRF on it) diagnoses the bath.

Hydrogen Embrittlement (HE): Atomic hydrogen soaking into steel during pickling/plating, making hardened steel **brittle** so it can crack later under load. ZnNi's low efficiency makes HE a real concern.

HE Relief Bake: Baking plated parts to drive hydrogen back out. For steel **> 40 HRC**: **375 ±25°F, ≥ 8 hr, within 4 hr of plating** (ASTM B850).

HRC (Rockwell C Hardness): A hardness scale. Above **40 HRC**, steel is high-strength and at real HE risk.

Inert (Insoluble) Anode: An anode that carries current but doesn't dissolve to feed metal — standard in alkaline ZnNi.

Inhibitor: An acid-pickle additive that slows attack on the base metal and reduces hydrogen absorption. Use on > 40 HRC steel.

Iron (Fe) Contamination: Dissolved iron dragged from the pickle. Causes haze/dark deposits. In alkaline ZnNi it can precipitate and be filtered (more forgiving than acid zinc), but still control it.

Membrane (Divided) Cell: A cell where the anode sits behind an ion-selective membrane, isolating anode reactions to protect the amine complexors and stabilize the alloy.

µm (Micron): Coating-thickness unit; 1 µm = 0.001 mm ≈ 0.04 mil. ZnNi typically 8–12 µm.

NaOH (Sodium Hydroxide): Caustic soda — dissolves zinc as zincate and carries current. ~100–140 g/L. Severe burn hazard.

Nickel (Ni): The alloying metal that gives ZnNi its corrosion and heat performance. Target **12–16% in the deposit.** A skin/respiratory sensitizer and regulated metal.

Nitric Acid (HNO_3): Strong oxidizing acid used (dilute) at Station 07 to brighten/activate the alloy before passivation. Severe hazard; never mix with caustic/organics.

Oxide: The thin film that forms on bare steel in air; blocks adhesion, so activation dissolves it right before plating.

Passivation (Passivate / Chromate): A trivalent-chromium conversion coating grown on the alloy that slows white rust and adds color; paired with a seal for top performance.

pH: A 0–14 acidity scale. The ZnNi plating bath is **13–14** (caustic); trivalent clear/blue passivate ~**1.6–2.5.**

ppm (Parts Per Million): Unit for small concentrations, used for contamination limits.

Rack Plating: Plating parts individually hung on a rack (vs. barrel). Common for critical ZnNi.

Rectifier: The power supply converting AC to the DC current that drives plating. High amperage — shock/arc-flash hazard. Always LOTO before servicing.

• GLOSSARY (CONTINUED)

RoHS: "Restriction of Hazardous Substances," banning certain materials (including Cr(VI)) in many products. Trivalent passivates comply; hexavalent does not.
Salt Spray Test: See ASTM B117. ZnNi with passivate + seal commonly reaches 720–1,000+ hr to red rust.
Seal / Topcoat: A layer applied over the passivate that boosts corrosion life and can control friction; standard on ZnNi to reach premium salt-spray numbers.
Sensitizer: A substance (like nickel) that can cause an allergic reaction that may become permanent with repeated exposure.
Service heat / Thermal Stability: ZnNi keeps protecting after heat exposure far better than plain zinc.
Sweet Spot (12–16% Ni): The deposit nickel window where ZnNi forms the protective gamma phase. More is NOT better.
Throwing Power / Distribution: The bath's ability to plate evenly into recesses. Alkaline ZnNi distributes thickness *and* alloy well.
Titrate / Titration: A chemical test measuring a solution's concentration (cleaner, bath components).
Trivalent: See Cr(III).
Turnover (Tank Turnover): One complete pass of the tank volume through the filter (or one full rinse-volume replacement).
Water Break Test: The free test for clean steel — water **sheets** off a clean part, **beads** on a dirty one.
White Rust: White powdery corrosion of zinc/zinc-nickel; the passivate slows it. (Red rust = the steel itself corroding through — the bigger failure.)
XRF (X-Ray Fluorescence): The non-destructive instrument that reads coating **thickness** and **%Ni** — the way you prove the alloy is in window.
Zincate: Zinc dissolved in caustic (the form zinc takes in an alkaline zinc/ZnNi bath).
Zn:Ni Ratio: The zinc-to-nickel ratio in the *bath* (~8:1 to 12:1) — the chemist's main lever for the deposit's %Ni.

★ **REMEMBER**

If you can confidently define **anomalous codeposition**, **the 12–16% Ni sweet spot**, **inert anodes / separate feed**, **amine complexor**, **XRF**, and **hydrogen embrittlement**, you understand what makes zinc-nickel different from every other zinc process. Those six terms are the spine of this manual.

☀ **FUN FACT**

Most of these terms are shared across all of metal finishing — learn them here and you can walk onto an acid-zinc, alkaline-zinc, or even a nickel or chrome line and follow the conversation. The vocabulary is the trade's common language.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

★ **REMEMBER**

Most zinc-nickel defects trace to **four root causes:** alloy out of window (temperature/ratio), contamination (Cr(VI) drag-back, iron, organics, carbonate), a rinse that let drag-in through, or poor prep. Diagnose with **XRF + Hull Cell** before you dump chemistry.

⚠ **HEADS-UP**

Before changing acid strength, pickle time, or anything that increases metal attack, check whether the part is **> 40 HRC** — those changes raise HE risk. When in doubt, ask your supervisor and lean on the relief bake.

STATION 01 — SOAK CLEAN

SYMPTOM	LIKELY CAUSE	FIX
Water break fails	Low cleaner conc. or temp	Titrate cleaner; raise temp
Oily film remains	Floating oil redepositing	Run skimmer; dump if heavily contaminated
White residue	Hard water / cleaner residue	Use DI rinse water
Etching/discoloration	Cleaner too aggressive/hot	Reduce temp; milder formulation
Spotty adhesion later	Uneven cleaning; air pockets	Reposition on rack; add agitation

STATION 02 — RINSE (PRE-CLEAN/ACTIVATION)

SYMPTOM	LIKELY CAUSE	FIX
Conductivity > 500 µS/cm	Too much drag-out; low flow	More drain time; more flow
Soapy residue	Inadequate rinse time	Extend dwell; add agitation
Next tank off-balance	Cleaner carry-over	Improve rinse; add a stage

STATION 03 — ELECTROCLEAN & ACID ACTIVATION

SYMPTOM	LIKELY CAUSE	FIX
Dark oxide remains	Acid dilute; dip too short	Increase conc.; extend time *(check HE first)*
Pitting/etching	Acid too strong/long/warm	Reduce conc.; shorten; add inhibitor
Adhesion fails later	Incomplete clean/activation	Re-check electroclean & acid; check alkaline drag-in
Excessive HCl fuming	High conc./temp; poor ventilation	Reduce; check exhaust
Dark smut	High Fe/Cu	Dump/recharge acid; find Cu source
HE failures	Over-pickle; cathodic clean on hard steel; no bake	Shorten; switch to anodic; inhibitor; bake (B850)

STATION 04 — RINSE (PRE-PLATE / GUARD)

SYMPTOM	LIKELY CAUSE	FIX
Conductivity > 200 µS/cm	Heavy acid drag-out; low flow	More drain off pickle; more flow
Bath caustic/ratio drifting	Acid drag-in	Tighten rinse; double rinse
Iron/filter load climbing	Fe from spent pickle	Refresh pickle; improve rinse
Parts oxidizing before plating	Slow transfer	Into plating within ~30 sec

STATION 05 — ZINC-NICKEL PLATING

SYMPTOM / READING	LIKELY CAUSE	FIX
%Ni below 12%	Low Ni in bath; temp low; CD high	Chemist: raise Ni/ratio; check temp/CD
%Ni above 16%	High Ni; temp high; CD low	Chemist: lower Ni; cool bath
Dull/dark deposit	Brightener low; organics; off-balance	Add brightener; carbon treat; rebalance
Burning at high CD	CD too high; metal off; poor agitation	Lower CD; rebalance; agitate
Bare/skip spots, blisters	**Cr(VI) drag-back**; poor cleaning	Stop drag-back; verify cleaning; chemist removes Cr
Hazy/rough	Iron/organics; carbonate high; filter	Analyze; carbon; remove carbonate; check filter
Peeling	Poor prep; oxidized before plating	Look upstream; speed handoff
Uneven %Ni across part	CD/agitation/temp variation	Check fixturing/agitation; XRF per spec

STATION 06 — RINSE (POST-PLATE)

SYMPTOM	LIKELY CAUSE	FIX
Finishing bath life short	Caustic/nickel drag-in	Tighten rinse; add recovery rinse
Blotchy passivate later	Residual caustic/nickel	Improve agitation/time
Nickel in effluent high	No recovery rinse	Add recovery; pump back to plating

STATION 07 — NITRIC/ACID ACTIVATION

SYMPTOM	LIKELY CAUSE	FIX
Blotchy passivate	Weak/skipped activation; caustic drag-in	Verify acid; improve Station 06
Thin alloy / over-etched	Over-dipped	Shorten time; lower conc.
Strong brown fumes	Over-concentrated; poor ventilation	Dilute to spec; check exhaust

STATION 08 — PASSIVATE (TRIVALENT)

SYMPTOM	LIKELY CAUSE	FIX
Thin/pale film	pH too high; bath depleted	Lower pH; replenish per TDS
Wrong color / weak black	pH off; wrong product; weak activation	Check pH; use ZnNi-specific; fix Station 07
Blotchy/uneven	Drag-in; uneven alloy	Improve rinse; check %Ni/uniformity
Film rubs off	pH off; not cured	Check pH; allow full cure

STATION 09 — SEAL / DRY

SYMPTOM	LIKELY CAUSE	FIX
Low salt-spray result	Seal skipped/under-cured; cured too hot	Apply/cure per TDS; respect temp limit
Cracked/hazy finish	Cured too hot	Lower cure temp to TDS limit
Marred film	Handled before cure	Wait full cure; clean gloves

ALLOY %NI QUICK-READ (XRF)

READING	WHAT IT MEANS / FIRST THING TO CHECK
12–16% Ni, even deposit	In window — you're good
Trending below 12%	Temp too low or Ni/ratio low — check temperature first
Trending above 16%	Temp too high or Ni too high — check temperature first
Edges high / recesses low (or vice-versa)	Current-density spread — check fixturing/agitation; read at spec spots
Sudden skip/bare with poor plating	Cr(VI) drag-back from passivate — stop the drag-back, call the chemist

💡 **TIP**

Notice how many finishing defects trace *upstream* to rinses, activation, and plating uniformity/%Ni. Most downstream defects are born upstream. Fix the root, not the symptom.

★ **REMEMBER**

The two instruments that solve most ZnNi mysteries are the **XRF** (is the alloy in window?) and the **Hull Cell** (is the bath bright and balanced across the CD range?). Run both before you change chemistry — diagnose first, adjust second, confirm third.

⚠ **HEADS-UP**

If a fix involves more acid, more time in the pickle, or anything that attacks the steel harder, **re-check the part's hardness** — a "quality fix" that raises hydrogen uptake on a > 40 HRC part is a safety problem. Lean on the relief bake and ask your lead.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ REMEMBER

An operator is "line-qualified" only when every station they run is signed. Nobody touches the **rectifier**, the **electroclean/pickle**, the **nitric activation**, or the **passivate** solo until the **Safety / PPE & SDS** row is signed. Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT
1	Safety / PPE & SDS (all stations)	____	________	________	____
2	Station 01 — Soak Clean + water break	____	________	________	____
3	Station 02 — Rinse (Pre-Clean) + conductivity	____	________	________	____
4	Station 03 — Electroclean & Acid Activate (incl. HE check)	____	________	________	____
5	Station 04 — Rinse (Pre-Plate / Guard)	____	________	________	____
6	Station 05 — Zinc-Nickel Plating operation	____	________	________	____
7	Alloy %Ni control — read XRF, hold 12–16% window	____	________	________	____
8	Hull Cell / bath test — run & interpret	____	________	________	____
9	Station 06 — Rinse (Post-Plate) + recovery rinse	____	________	________	____
10	HE Bake (ASTM B850) decision & logging	____	________	________	____
11	Station 07 — Nitric/Acid Activation	____	________	________	____
12	Station 08 — Passivate (trivalent chromate)	____	________	________	____
13	Station 09 — Seal/Topcoat, Final Rinse/Dry & cure	____	________	________	____
14	LOTO on rectifier	____	________	________	____
15	Emergency response — eyewash, shower, spill	____	________	________	____
16	Waste / drag-out / nickel recovery handling	____	________	________	____
17	Troubleshooting — symptom & %Ni recognition	____	________	________	____

Operator name: ______________________________ **Hire/start date:** __________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ____________

⚠ HEADS-UP

Safety rows (1, 14, 15) are not optional and not "fill in later." An operator may not work *any* station until PPE/SDS, LOTO, and emergency response are signed off. Safety competency comes before production competency, always.

💡 TIP

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell (e.g., "Q 6/26 · recert 6/27"). A yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS · PASSING SCORE 80% (16/20)

ZINC-NICKEL COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ______________ **Score:** ________ / 20 **Safety-gate Qs (all must be correct): 5, 6, 14, 19**

Q1. Zinc-nickel's headline advantage over plain zinc is:
A. It plates faster B. It costs less C. Far better corrosion protection (e.g., 720–1,000+ hr salt spray with passivate + seal) D. It needs no passivate

Q2. (Short answer) What is the target **nickel content in the deposit**, and why is "more nickel" *not* better?

Q3. The alkaline zinc-nickel plating bath operates at about what pH?
A. 1.5–2.2 B. 4.8–5.8 C. 7.0 (neutral) D. 13–14

Q4. (Short answer) What does "**anomalous codeposition**" mean, and why does it mean the bath's Zn:Ni ratio is *not* the same number as the deposit's %Ni?

Q5. [SAFETY GATE] Zinc-nickel is a low-efficiency, high-hydrogen process. For high-strength steel **> 40 HRC**, the relief bake (ASTM B850) is:
A. 200°F for 1 hour, any time that week B. 375 ±25°F for at least 8 hours, started within 4 hours of plating C. 500°F for 30 minutes D. No bake needed

Q6. [SAFETY GATE] Nickel in this bath and its mist is primarily a:
A. Harmless additive B. Skin and respiratory **sensitizer** and regulated heavy metal C. Flammable gas D. Strong base only

Q7. (Short answer) Name the **instrument** used to measure the actual %Ni (and thickness) of the deposit, and why bath chemistry alone isn't enough to certify a zinc-nickel job.

Q8. Why are the anodes in alkaline zinc-nickel usually **inert (insoluble)** instead of dissolving like acid-zinc anodes?
A. There's no anode that dissolves in the right 12–16% proportion, so metals are fed separately and the alloy is actively controlled B. They're cheaper to scrap C. They plate faster D. To raise pH

Q9. (Short answer) Which single operating parameter most often **drives %Ni drift** day-to-day, and which way does %Ni tend to move if it climbs?

Q10. Alkaline zinc-nickel's cathode efficiency is about:
A. ~50–65% (low) B. 95–98% C. 100% D. 80–90%

Q11. (Short answer) What is the job of the **amine complexors** in the bath, and why can't nickel just be added to a caustic bath without them?

Q12. A part comes out with **bare/skip spots and blisters**, and the bath suddenly plates poorly. The classic cause on a zinc-nickel line is:
A. Too much boric acid   B. Anodes too pure   C. Water too pure   D. **Hexavalent chromium (Cr(VI)) dragged back** from the passivate into the plating bath

Q13. Typical commercial zinc-nickel coating thickness is about:
A. 0.5–1 µm   B. 8–12 µm   C. 100–150 µm   D. 2 mm

Q14. (Short answer) **[SAFETY GATE]** Name the special hazards of the **nitric acid** activation (Station 07), including one chemistry it must **never** be mixed with, and the universal acid-dilution rule.

Q15. Why is the **post-plate rinse (Station 06)** often run as a still "recovery" rinse?
A. To recover expensive **nickel** (pump it back to plating) and reduce waste   B. To warm the parts   C. To add brightener   D. It isn't useful

Q16. Modern RoHS-compliant zinc-nickel passivation uses which chromium?
A. Hexavalent — Cr(VI)   B. Cyanide   C. Trivalent — Cr(III)   D. None

Q17. (Short answer) What does the **seal/topcoat (Station 09)** do for zinc-nickel, and why is it usually required to hit the high salt-spray numbers?

Q18. The protective zinc-nickel structure formed in the 12–16% Ni window is called the:
A. Alpha phase   B. Rust layer   C. **Gamma (γ) phase**   D. Zincate phase

> ⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
> Questions 5, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) **[SAFETY GATE]** List **four** pieces of PPE you must wear at the zinc-nickel **plating** station (which has caustic + nickel + amine mist), and name the procedure required before any **rectifier** maintenance.

Q20. (Short answer) Zinc-nickel costs more and plates slower than plain zinc. Name **two** reasons a shop runs it anyway (i.e., what zinc-nickel gives you that plain zinc can't).

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training before sign-off, regardless of total score.

Q1. C — Far better corrosion protection (720–1,000+ hr salt spray with passivate + seal). That's the whole reason to run the more expensive, slower process.

Q2. Target **~12–16% nickel** (bullseye ~13–15%). More is not better because **above ~16–18% the alloy becomes less sacrificial** (may stop protecting the steel), gains stress/cracking, and passivates poorly; below ~12% it loses the corrosion benefit. There's a **sweet spot**, not a "more is better" dial.

Q3. D — 13–14 (strongly caustic).

Q4. Anomalous codeposition means the **less noble metal (zinc) plates preferentially**, so the deposit comes out far lower in nickel than the bath. A bath running ~8:1–12:1 Zn:Ni yields only **12–16% Ni in the deposit** — so the bath ratio and the deposit %Ni are different numbers, and you must **measure %Ni** rather than assume it.

Q5. B — 375 ±25°F for at least 8 hours, started within 4 hours of plating (ASTM B850).

Q6. B — A skin and respiratory **sensitizer** and a regulated heavy metal. (Don't breathe the mist; keep it off skin.)

Q7. XRF (X-Ray Fluorescence) — non-destructive, reads thickness and %Ni in seconds. Bath chemistry only *predicts* the alloy; **XRF proves** it's in the 12–16% window, which is the spec the customer buys.

Q8. A — There's no anode that dissolves in the right 12–16% proportion; inert anodes carry current while zinc and nickel are fed *separately* and the alloy is actively controlled (some lines add a membrane to protect the complexors).

Q9. Temperature is the biggest everyday driver of %Ni drift. If temperature **climbs, %Ni tends to climb** (often out the top of the window). Check temperature first when %Ni drifts with no chemistry change.

Q10. A — ~50–65% (low), which is why ZnNi plates slowly and why hard parts must be baked.

Q11. The amine complexors **wrap around the nickel and hold it dissolved** at high pH (and control how much codeposits). Without them, nickel would **precipitate out as useless sludge** in the caustic bath.

Q12. D — Hexavalent chromium (Cr(VI)) dragged *back* from the passivate into the plating bath — causes skip/bare spots and blisters and is hard to clear. Prevent with good rinsing and correct flow.

Q13. B — 8–12 µm (severe service up to ~20 µm).

Q14. Nitric is a strong oxidizer with **severe burns and irritating brown (nitrogen-oxide) fumes**; **never mix it with the caustic bath / cyanide / organics** (violent or toxic reaction). **Always add acid to water, never water to acid.** (Accept any reasonable mix of these.)

Q15. A — To recover expensive **nickel** (pump the recovery-rinse water back to the plating bath) and reduce waste-treatment load.

Q16. C — Trivalent, Cr(III). (Hexavalent is a known carcinogen and not RoHS-compliant; some legacy ZnNi blacks were hexavalent — confirm yours is trivalent.)

Q17. The seal/topcoat adds a **barrier layer over the passivate** (and often controls fastener friction), pushing corrosion life to **720–1,000+ hours** to red rust. Passivate alone usually can't reach those numbers — the seal is how ZnNi earns its premium spec.

Q18. C — The gamma (γ) phase.

Q19. PPE (any four): sealed chemical splash goggles (*not* safety glasses), face shield, elbow-length chemical-resistant gloves, chemical apron, chemical-resistant boots, **respirator** (nickel/amine mist). **Rectifier procedure: LOTO (Lockout/Tagout)** — de-energize and lock out before any maintenance.

Q20. Any two of: **outstanding corrosion protection** (720–1,000+ hr); **thermal stability** (holds protection after heat/baking); **harder, more wear-resistant deposit**; **uniform alloy & thickness even in recesses**; **survives the embrittlement bake** without losing corrosion life. (Plain zinc can't match these.)

Answer distribution (MC): A=3 · B=3 · C=3 · D=2

★ **REMEMBER**

A score of 16/20 passes — but every safety question (5, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

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CERTIFICATE OF COMPLETION

ZINC-NICKEL ALLOY PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Zinc-Nickel Alloy Plating** training course — covering all nine process stages (Clean, Rinse, Electroclean & Activate, Rinse, Zinc-Nickel Plate, Rinse, Nitric Activate, Passivate, and Seal/Final Rinse/Dry), alloy codeposition and the 12–16% nickel sweet spot, %Ni control and XRF measurement, bath chemistry and maintenance (caustic, complexed nickel, inert anodes/membrane), Hull Cell diagnostics, hydrogen-embrittlement control and baking, troubleshooting, and the safety practices for hot caustic cleaners, mineral and nitric acids, nickel and amine complexors, the zinc-nickel bath, and chromate passivation — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ________________ **Re-certification due:** ________________

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR